EnergyMind Tunisia

Phase 1 Project Proposal

Trustworthy monitoring of a smart-metered neighbourhood: telling grid events from bad data

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Abstract

A smart meter that reports something surprising is telling one of two stories: either the *grid* did something unusual (an outage, a voltage sag, a demand surge) or the *data* is wrong (a drifting sensor, a corrupted value, a bypassed or falsified meter). The two need opposite responses, and confusing them is costly. We propose a human–AI system for a smart-metered low-voltage neighbourhood that (1) detects anomalous and biased readings with a hybrid of physics rules, statistics and unsupervised learning, (2) explains each verdict with a calibrated confidence and an auditable record, (3) routes every case to the right level of human control, never letting confidence override consequence, and (4) lets several neighbourhoods learn from each other without sharing raw readings. Phase 1 delivers a labelled, Tunisia-calibrated simulated dataset, the design of all four goals, and a first detection prototype run on one neighbourhood: at 2.3% false positives it detects 30 of 36 genuine events, 23 of 30 meter faults and 6 of 8 falsifications, and it shows plainly which cases remain hard.

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1 Scope

The slice of the energy system we monitor. The low-voltage distribution network of a smart-metered residential neighbourhood, from the secondary of the MV/LV transformer to the customer meters:

- **Customer smart meters** reporting every 30 minutes: imported energy, PV export, reactive energy, mean/min/max voltage, current, power factor, voltage distortion and an event log.
- **One accurate meter at the head of each feeder**, which sees the true energy flow and the feeder voltage. It makes two physical cross-checks possible: the feeder energy balance (head minus the sum of customer meters) and the comparison of each meter's voltage with the head's.
- **Context**: weather (temperature, irradiance), calendar (weekends, Tunisian public holidays, Ramadan) and the static description of each meter and home.

Single-node prototype and multi-node design. Our simulated dataset contains three such neighbourhoods (*nodes*) with different climates and customer mixes (Table 1). As the challenge requires, Goals 1–3 are prototyped on a single node, **N1** (coastal suburb: one transformer, 30 meters on three feeders of 10, three homes with rooftop PV). Goal 4 connects N1, N2 and N3; multi-node sharing is a documented design in Phase 1, as the challenge allows.

Table 1. Nodes of the simulated dataset (60 meters, 7 feeders, year 2025).

Node	Profile	Meters	Feeders	Role
N1	Coastal suburb	30	3	Phase 1 prototype (Goals 1–3); sharing participant
N2	Hot interior town	15	2	Sharing participant (Goal 4)
N3	Mixed residential / commercial	15	2	Sharing participant (Goal 4)

The question the system answers. For every suspicious reading: *can this data be trusted, and if not, why, how sure are we, and who should act?* The answer is one of four verdicts shared by all components — `NORMAL`, `GENUINE_EVENT` (the data is right and the grid did something), `METER_FAULT` (a faulty or corrupted meter) and `SUSPICIOUS_DATA` (readings that look manipulated) — plus `NO_DATA` when readings are missing.

Out of scope. Control of real STEG equipment (the system recommends; switching and disconnection always need human approval); the medium- and high-voltage network; real STEG meter data (all data are simulated and labelled as such); billing and customer-facing processes.

2 Objectives

Each objective is tied to a measurable target. Phase 1 targets are design deliverables and a first measured baseline; Phase 2 targets are the ones we will report on the full prototype.

Table 2. Objectives and targets.

Objective	Phase 1 (this submission)	Phase 2 target (measured on N1)
O1 Detect anomalous and biased readings (Goal 1)	Hybrid detector designed and run on N1: 2.3% false positives on clean meter-days; 30/36 genuine events, 23/30 meter faults, 6/8 falsifications detected	Keep false positives $\leq 2.5\%$; detect the physically consistent cases (bypass, bias, drift) now missed, via a modelled feeder-balance residual
O2 Explain each verdict (Goal 2)	Rule cascade, 4-class calibrated confidence, explanation pipeline and log schema designed	Brier score and calibration curve per class; $\geq 90\%$ of named channels and scopes correct on labelled events; faithfulness check on every LLM text
O3 Govern human–AI decisions (Goal 3)	Governance levels, risk model, routing rules, blind-first console and scenarios designed	Zero unsafe automatic actions; missed-escalation and false-escalation rates against the dataset's reference responses; coverage–accuracy curve
O4 Share knowledge across nodes (Goal 4)	Four protected channels, privacy accounting, threat model and canary adoption designed	Detection of attack types a node has never seen, with vs. without sharing; privacy–utility curve; corrupted-node test
O5 Be honest about evidence	Every measured number is printed by code in the repository (`scripts/check_dataset.py` for the data, `docs/goal1_detection/n1_experiment.py` for detection; the governance scenarios and safety invariants are tests in `tests/test_governance.py`); untested items are labelled as design	Results reported per event with confidence intervals, never as a single accuracy figure

3 Data and Anomaly-Simulation Plan

Real fault- and attack-labelled distribution data is rare and is not public for Tunisia, so the project uses a simulated dataset with exact labels. Full documentation is in the repository's data card (`docs/data_card.md`).

3.1 The dataset

- **One table, one row per meter per half-hour**: 1 051 200 rows × 67 columns; 60 meters on 7 feeders in 3 nodes; 1 January–31 December 2025, local time. Designed and generated by our team with a fixed random seed (2025); the generator in `simulate/` (healthy network, anomaly injection, join) reproduces the committed file exactly, and every figure quoted from it is printed by `scripts/check_dataset.py`.
- **Column groups**: identifiers; meter measurements (10); meter and home details (21); weather and calendar (6); feeder context (11, including feeder-head energy and voltage and the energy-balance gap); ground truth (15 columns prefixed `gt_`).
- **Label hygiene**: the `gt_*` columns and two simulator settings are never model inputs; five "oracle" columns (true energy, true voltage, true losses) are used only to analyse errors.

3.2 How the normal behaviour is generated

1. **Weather** for three Tunisian climates (coastal, interior, southern coast), with heat waves and solar irradiance.
2. **Household load** with Saturday–Sunday weekends, public holidays, **Ramadan** (suhoor and iftar peaks), summer vacations, temperature-driven air conditioning, electric heating, and rooftop PV with export.
3. **Calibration** to public Tunisian figures: residential consumption near the STEG average of about 1 450–1 470 kWh per year (N1: 1 475 kWh); homes with air conditioning use about 1.68× more in July–August than in January–February.
4. **Electrical model**: voltage drops along each feeder and rises with PV export; power factor falls with air-conditioning load; meters add small measurement noise (energy agrees with $V\,I\,\mathrm{PF}\,\Delta t$ within 0.93% median, 2.7% at the 95th percentile).
5. **Feeder-head meter** seeing true energy plus line losses and unmetered street lighting, which creates the realistic $\approx 5\%$ feeder balance gap.

3.3 How anomalies are injected

Genuine events are injected first, then data problems and falsification at meter level. Every row is labelled with its class (`gt_label`), exact cause (`gt_subtype`, 29 values), event identifier (`gt_event_id`), whether the data can be trusted, and a suggested response. Rows are labelled only while the anomaly is visible (a drift once it reaches 1%, a selective bypass only in the bypassed hours).

Table 3. Anomaly classes and injection recipes. 93.2% of rows are normal.

Class	Data trusted?	Injected causes (recipe)	Rows
`grid_event`	yes, genuine	outage and post-restoration surge; scheduled load shedding (July–August); voltage sag (−15 to −45 V) and swell; demand surge (×1.35–1.8 for 1.5–4 h); cold-load pickup	0.37%
`equipment_failure`	yes, genuine	cable fault (meters beyond the cut lose supply); transformer degradation ending in a trip	0.64%
`sensor_fault`	no	gain drift (slow, mostly under-reading); bias step (×0.86–0.93 or ×1.07–1.14 for 3–11 weeks); voltage-sensor offset drift; clock shift (1–3 h); stuck value (3–48 h)	2.90%
`data_corruption`	no	spike (×8–40); sign flip; saturation (99.999); Wh reported as kWh	0.02%
`falsification`	no	*physical*: partial bypass (energy and current ×0.4–0.8), selective bypass, meter zeroing; *data-only*: scaling (×0.3–0.75), evasive random-walk scaling, replay of last week, inflation (framing a neighbour), coordinated attack on three meters	2.74%
`comm_loss`	no data	all measurements missing for 1.5–48 h (PLC meters most exposed)	0.10%

Why detection is deliberately not trivial. A real event hits neighbours together, a failing sensor is isolated — except for the coordinated attack. Physical tampering keeps energy, voltage and current consistent and only opens the feeder balance; data-only injection breaks that consistency but leaves voltage untouched. Hard negatives are included on purpose: summer vacations (genuinely low use), the Ramadan shift, heat-wave surges, PV export reversing the current, and post-outage surges.

3.4 What the data looks like on the prototype node

Node N1 has 98 labelled events in 2025: 37 genuine (33 grid events, 4 equipment failures), 33 meter faults (25 corruption, 8 sensor faults), 8 falsifications (one of each type) and 20 communication losses. Two facts shape the whole design:

- **A supply loss is recorded as zeros, not as missing data** (683 readings in N1), so "flat at zero" must be read as an outage before it is read as a frozen meter.
- **Rare events are really rare**: each falsification type occurs once in N1 and 2–3 times in the whole dataset, at most once per node.

3.5 Evaluation protocol

A calendar split (train January–August, validate September–October, test November–December) does not work here: in N1 the last two months contain no falsification and no sensor-bias event. All goals therefore use **5-fold cross-validation with folds grouped by event** (no event is split between fitting and testing; clean days grouped by meter-week), unsupervised models are trained on clean days only, and results are reported per class and per event with intervals. This protocol is implemented in the repository (`energymind.data.event_folds`).

3.6 Known limitations of the data

The data are synthetic: a detector tuned here may overfit the generator. The anomaly share ($\approx 6.8\%$) is higher than in reality so that models have examples. Meter fields follow an assumed IEC-style half-hourly profile. There is one transformer per node and no three-phase imbalance. All results are reported as simulated.

4 Preliminary Design: Overview

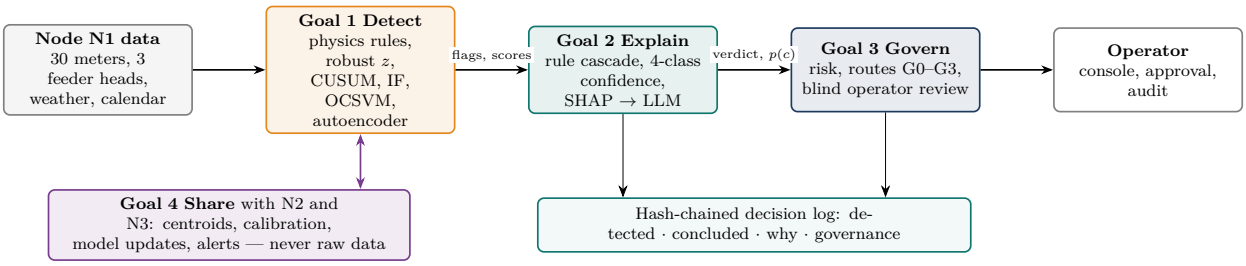


Figure 1. System overview on one node. Sections 5–8 give the preliminary design of each goal; Goal 3 is condensed here, its complete design is in `docs/goal3_governance`.

5 Goal 1: Anomaly and Bias Detection

Summary. Our system must decide, for every suspicious reading, whether the *grid* did something unusual or the *data* is wrong, because the two need opposite responses. No single detector can do this. We therefore combine (i) physics and statistical rules, which are auditable, (ii) three unsupervised machine-learning detectors trained only on clean data, which are sensitive to subtle patterns, and (iii) a cross-sensor consistency reasoner that separates genuine events from data-quality problems. Section 5.5 tests the detection layers on node N1 of the EnergyMind dataset and reports, per fault type, what they catch and what they still miss.

5.1 The Problem: a Suspicious Number Has Two Possible Meanings

An anomaly is a pattern that deviates from expected behaviour [1]. In our prototype node (N1 of the EnergyMind dataset: one transformer, 30 smart meters on 3 low-voltage feeders of 10, three homes with rooftop PV, readings every 30 minutes, each feeder metered at its head) a surprising value can come from the **grid** or from the **data chain**, and the right response differs: a real event needs operational action, a faulty or falsified sensor needs to be distrusted and repaired. A further difficulty is that an informed attacker can craft false data that passes standard bad-data checks [3], so one test is never enough.

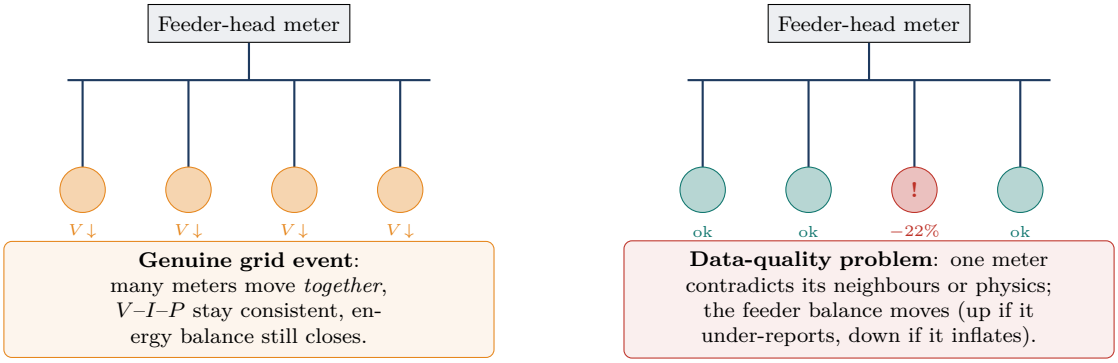


Figure 2. The core question. Left: a real event is visible on many sensors and obeys physics. Right: a faulty or falsified channel disagrees with redundant measurements. This cross-sensor logic is the backbone of our design.

5.2 Types of Anomalies

The literature classifies anomalies by how they relate to context [1]. We use that classification and map it onto grid failures (Figure 3, Table 4).

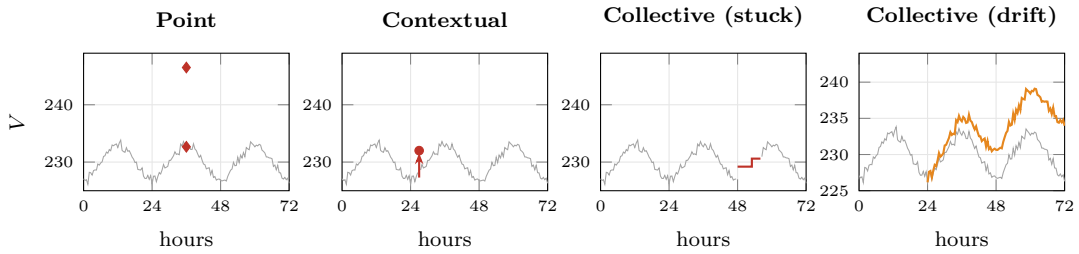


Figure 3. Illustrative voltage traces (synthetic, 30-min samples as in the dataset). Grey: normal behaviour. Red/amber: the anomaly. A *point* anomaly is one odd value; a *contextual* anomaly is a value that is normal in general but wrong for its time of day; *collective* anomalies are odd only as a sequence (a frozen sensor, a slow drift).

Table 4. From anomaly types to the fault types of the EnergyMind dataset. Genuine events are *expected* to trigger the detectors; the reasoner (Section 5.5) must not call them bad data. Missing readings (communication loss) are not scored: they are passed on as "no data".

Project anomaly	Type [1]	Signature in the data	First line of defence
Impossible value (spike, sign or unit error, saturation)	Point	$E < 0$, $E = 99.999$, $\times 1000$, $\times 8$–40	Range rules
Stuck meter	Collective	Every channel frozen (E *and* V)	Flat-line rule
Data-only injection (scale, evasion, inflate, coordinated)	Contextual	$E \neq VI \cos\varphi \, \Delta t$	Physics rule (skipped while exporting)
Replay	Collective	Same half-hour one week earlier	Profile residual, ML ensemble
Voltage sensor offset	Collective	Meter V persistently *above* feeder-head V	V-vs-head rule, CUSUM
Sensor bias, gain drift (± 2–14%)	Collective	V, I, E scaled together: physics holds	Feeder balance (feeder level), profile, ML
Physical bypass, zeroing	Contextual	Energy *and* current reduced: physics holds	Feeder balance, zero-run, ML
Clock shift	Collective	All channels delayed 1–3 h	Lag correlation with profile
Genuine outage, sag, swell, surge, cable fault	(not an error)	Many meters, physics holds; $V = 0$ on supply loss	Flagged here; classified by the reasoner

5.3 Families of Detection Methods and Our Choice

Methods can be organised by *what knowledge they use* and by *how many labels they need* [1, 2]. Because labelled faults are rare in grids, we are driven toward the semi-supervised and unsupervised regimes: learn what "normal" looks like from clean data, and flag deviations (Figure 4).

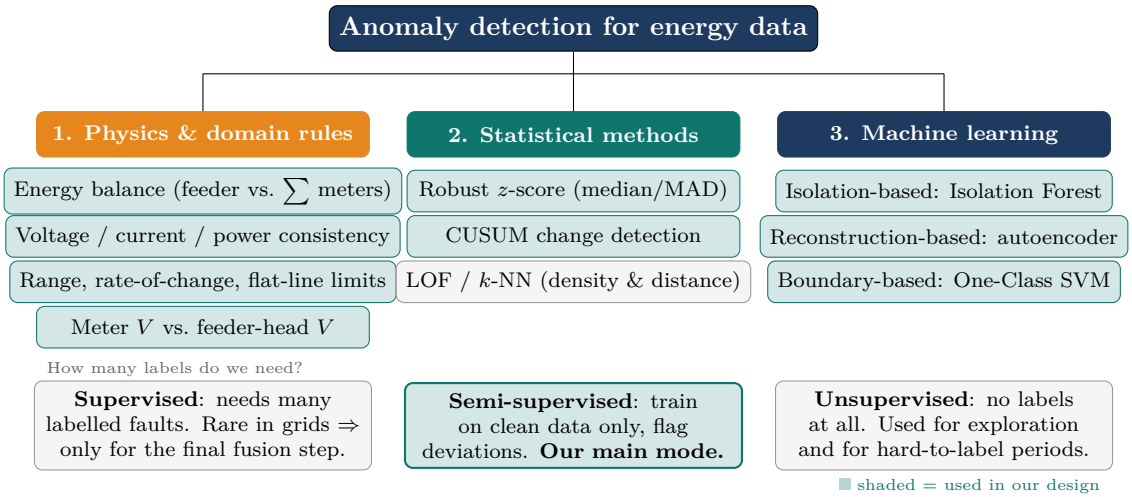


Figure 4. Taxonomy of detectors. Teal boxes are the methods we use (detailed in Section 5.4). Hybrid physics + learning follows the argument that high-stakes decisions should rest on inspectable models where possible [4].

5.4 The Methods, One by One

5.4.1 Statistical Outlier Methods

Robust z-score. The classic $z = (x - \bar{x})/s$ is distorted by the very outliers it seeks, because mean and standard deviation have a 0% breakdown point. Replacing them by the median and the scaled median absolute deviation gives

$$z_i^{\text{rob}} = \frac{x_i - \text{med}(x)}{1.4826 \cdot \text{MAD}(x)}, \qquad \text{MAD} = \text{med}_i |x_i - \text{med}(x)|, \tag{1}$$

which tolerates up to 50% contamination [5, 6]. **CUSUM** [7] accumulates small deviations, $S_t = \max(0,\, S_{t-1} + (x_t - \mu_0)/\sigma - k)$, and alarms when $S_t > h$; it is the natural tool for slow drift. **LOF** [8] compares a point's local density with that of its neighbours, so it handles clusters of different density.

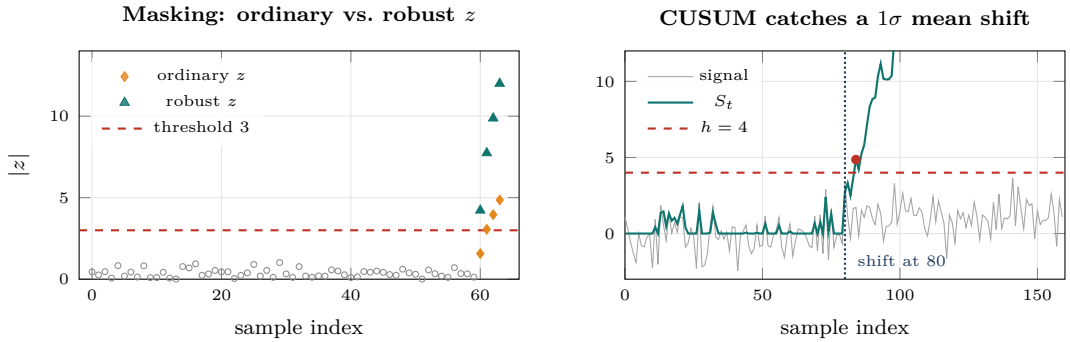


Figure 5. Left: four injected outliers among 60 normal samples. Ordinary z gives 1.6, 3.1, 4.0, 4.9, so 1 of them hides below the threshold (masking); robust z gives 4.2, 7.7, 9.9, 12.0, so all four are flagged. Right: CUSUM ($k = 0.5$, $h = 4$) alarms at sample 84, 4 samples after the shift, although the raw signal stays mostly within $\pm 3\sigma$. Both from our `gen.py`, synthetic data, fixed seed.

Role in our design. Cheap, transparent first line and the baseline every ML component must beat. Robust z is applied to residuals from each meter's own 28-day half-hourly profile, so the evening peak, weekends, Ramadan and heat-wave days are not mistaken for anomalies. CUSUM runs on two slow signals: the meter's voltage relative to its feeder head (a sensor offset) and the feeder's daily energy-balance gap (a bias or bypass somewhere on the feeder).

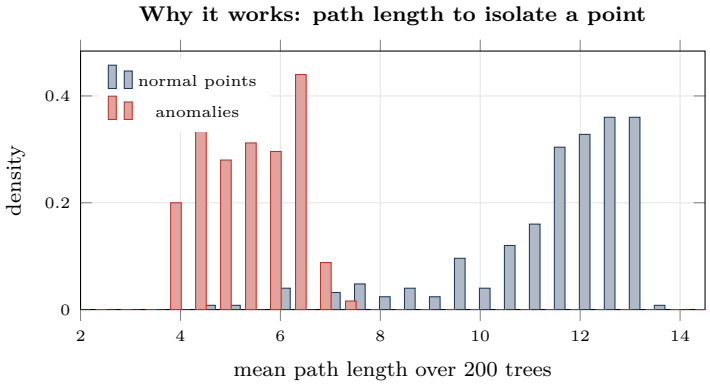
The physics rules and what they cannot see. The rule $E \approx V I \cos\varphi\, \Delta t$ exposes *data-only* injection, which edits the energy field and leaves V and I untouched. It is skipped while a meter exports PV energy (import is then 0 while current flows; without the gate it fires on 39% of the clean readings of N1's PV homes). It is *blind* to sensor bias, gain drift and physical bypass, which scale energy and current together, so those cases rest on the feeder balance and on the meter's own history. A meter cannot read a higher voltage than the head of its feeder, because voltage only drops along the line; a persistent excess (24-h median above +1.5 V, export readings excluded) is a voltage-sensor offset.

5.4.2 Isolation Forest

Liu, Ting and Zhou [9] observed that anomalies are few and different, so random axis-parallel cuts isolate them in fewer splits. A tree is grown by repeatedly picking a random feature and a random split value; the path length $h(x)$ to isolate x is short for anomalies. With subsample size ψ the score is

$$s(x, \psi) = 2^{-E[h(x)]/c(\psi)}, \qquad c(\psi) = 2H(\psi - 1) - \frac{2(\psi - 1)}{\psi}, \qquad H(i) \approx \ln i + 0.5772, \tag{2}$$

so $s \to 1$ for anomalies and $s < 0.5$ for normal points. It is linear in time, needs no labels and no distance computations. The Extended Isolation Forest uses randomly oriented cuts and removes the axis-parallel artefacts visible in score maps [10], which matters for correlated quantities such as V and I.



Reading the plot. Normal points sit deep inside dense regions, so many random cuts are needed to separate them (mean $\approx$ 11.6). Anomalies are isolated after only 5.2 cuts on average. A short average path means a high score s. Same kind of 2-D cloud as Figure 7.

Figure 6. Average isolation path length from the fitted Isolation Forest (200 trees; 250 normal points and 250 anomalies drawn outside the dense core).

5.4.3 One-Class SVM

Schölkopf et al. [11] separate the data from the origin in a kernel feature space with maximum margin:

$$\min_{w,\xi,\rho} \ \tfrac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho \quad \text{s.t.} \quad \langle w, \phi(x_i)\rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \tag{3}$$

and $f(x) = \operatorname{sgn}\big(\langle w, \phi(x)\rangle - \rho\big)$ with an RBF kernel. The parameter ν upper-bounds the fraction of training points treated as outliers. The closely related SVDD [12] encloses the data in a minimal hypersphere.

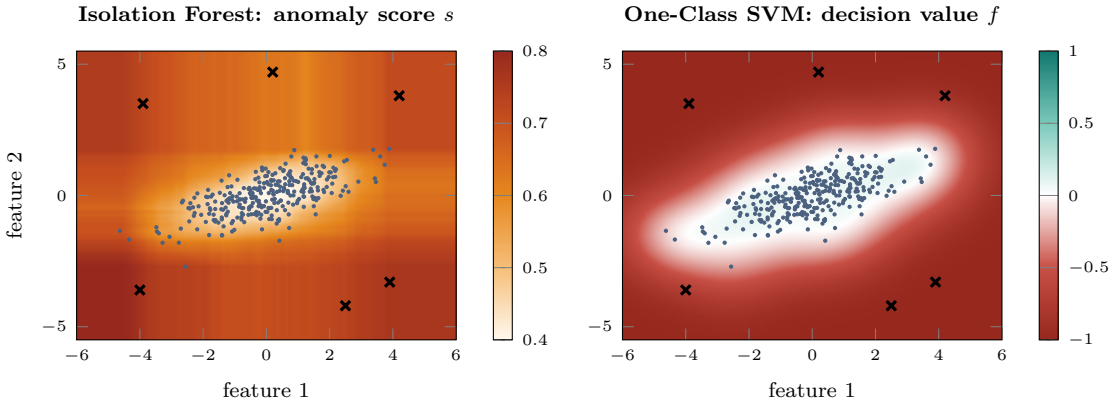


Figure 7. Two detectors fitted by `scikit-learn` [16] on the same correlated 2-D "normal" cloud (blue dots); crosses are injected outliers. Isolation Forest (200 trees, $\psi = 256$) scores the outliers 0.63–0.77 against a mean of 0.46 for normal points. The One-Class SVM ($\nu = 0.05$, RBF; decision value scaled to $[-1, 1]$) draws a smooth boundary (white) around the data. Their different inductive biases are why disagreement carries information.

Role in our design. Both run on daily window features of each meter (48 readings): profile residual (mean, maximum, spread), change of the meter's share against its feeder neighbours, physics ratio, V minus feeder-head V, share of zero readings, longest frozen run, lag gain against the profile (clock shift), similarity to the same day a week earlier (replay), distortion and minimum/maximum voltage. Isolation Forest is our fast first ML pass. OCSVM is trained on a subsample (it scales

poorly) as a second, differently-biased opinion. When they disagree, confidence drops and the case is routed to a human.

5.4.4 Autoencoders

An autoencoder learns to compress and reconstruct normal data through a narrow code. Trained on clean data only, it reconstructs anomalies badly, so the reconstruction error is the anomaly score [13]:

$$e(x) = \|x - g(f(x))\|^2 = \sum_j e_j, \qquad e_j = (x_j - \hat{x}_j)^2. \tag{4}$$

LSTM encoder–decoders extend this to multi-sensor time series [14]; surveys of deep detectors discuss when they help and where they over-generalise and reconstruct anomalies too well [2]. Crucially for us, the *per-channel* error e_j points at *which sensor* looks wrong, and it can be fed to Shapley-value explanations [15].

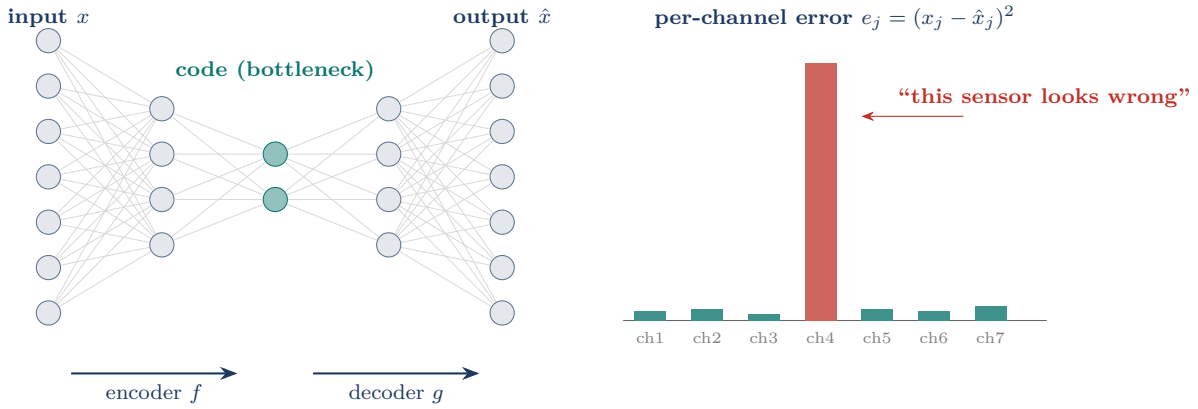


Figure 8. Autoencoder principle (left) and the per-channel reconstruction error that localises the faulty sensor (right; illustrative values).

Role in our design. Trained on clean windows only; captures correlated-sensor inconsistencies and subtle waveform changes that thresholds miss, and gives channel-level evidence for the explanation sentence ("channel 4 deviates most").

5.4.5 The Methods at a Glance

Table 5. Qualitative comparison, our assessment from the cited literature [1, 2] and the experiment in Section 5.5.1. Green = favourable, amber = acceptable, red = costly.

Method	Labels	Run cost	Explainable	Strong on	Weak on
Rules / physics	none	very low	by design	Impossible values, data-only injection, voltage offset	Bias and bypass (physics still holds)
Robust z / CUSUM	none	very low	high	Spikes; persistent offsets	Seasonal baselines (feeder balance)
Isolation Forest	none	low	medium (TreeSHAP)	Global outliers, many features	Shifts inside the normal range
One-Class SVM	none	high on big data	low (post-hoc)	Smooth boundary on compact data	Scale; kernel tuning
Autoencoder	none	medium (training)	medium (per channel)	Correlated-sensor, waveform change	Can reconstruct anomalies too well

5.5 Putting It Together: Our Hybrid Detection Stack

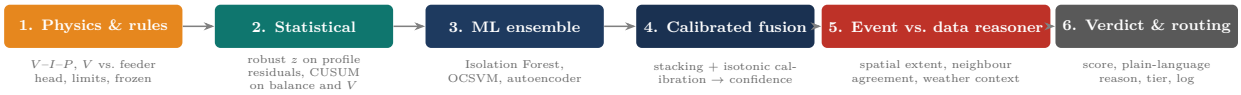


Figure 9. The detection pipeline. Rules can veto or force escalation; ML scores add sensitivity; the reasoner decides *what kind* of unusual it is.

How the verdict is reached. Layers 1–3 answer "is something unusual?" and are *meant* to fire on genuine events too; layer 5 answers "what kind?" from consistency features (Table 6). Layers 5–6 are specified in Sections 6 and 7; this section covers layers 1–4.

Table 6. Consistency features used by the event-versus-data reasoner.

Evidence	Genuine event	Data-quality problem
Spatial extent (meters affected)	Many	One or a few
Physical consistency (V, I, P)	Holds	Violated (data-only injection) or holds (bias, bypass)
Feeder energy balance	Unchanged	Opens (under-reporting) or goes negative (inflation)
Supply	$V = 0$ on many meters (outage)	$V > 0$
Agreement with correlated neighbours	Yes	Contradicts them
Temporal shape	Physical dynamics	Step, frozen, replayed
Weather / time-of-day context	Explains it	Does not explain it

5.5.1 Results on Node N1 of the Challenge Dataset

Set-up. Unit of analysis: one meter-day (48 readings) of node N1, from 8 January (the profiles need one week of history) to 31 December, excluding days that are mostly missing: 10,715 meter-days, of which 9,494 are clean and the rest belong to 74 labelled events. Every detector is trained on clean meter-days only; labels are used only to score. Thresholds give a 1% false-positive rate on held-out clean days (the stack: rules, or any learned detector at 0.25%). Everything is out-of-fold over 5 folds, with clean days grouped by meter-week and anomalous days grouped by event, so no event is split. A time split is not possible: in N1, November–December contain no falsification and no sensor-bias event. An event counts as *detected* when at least 20% of its meter-days are flagged (any flag for events of five days or less), so a long event is not "caught" by a chance alarm. Code: `n1_experiment.py`.

Table 7. Node N1, out-of-fold. "days": share (%) of the class's meter-days flagged (darker = higher). "events": events detected / events. False positives: share of clean meter-days flagged. Genuine events: outage, load shedding, sag, swell, surge, cable fault, transformer; meter faults: corruption and sensor faults; falsification: the 8 manipulation types. Simulated data.

	Genuine event		Meter fault		Falsification		False
Detector	days	events	days	events	days	events	**positives**
Rules (layer 1)	42	17/36	29	18/30	66	5/8	0.0%
Robust z	1	3/36	26	12/30	61	5/8	1.1%
CUSUM	1	4/36	12	2/30	6	0/8	2.1%
Isolation Forest	3	11/36	7	15/30	9	2/8	0.9%
One-Class SVM	74	30/36	31	25/30	67	6/8	1.3%
Autoencoder	74	30/36	28	22/30	64	5/8	1.1%
Detection stack	74	30/36	31	23/30	69	6/8	2.3%

Table 8. The 16 falsification and sensor-fault events of N1, one by one (most types occur once, so percentages per type would be meaningless). R = rules, z = robust z, C = CUSUM, IF, SVM, AE = autoencoder. First flag and days flagged refer to the whole stack. Balance alarm: first day the CUSUM on the feeder's daily energy-balance gap fires (it is in alarm on 5.2% of clean feeder-days); it names a feeder, not a meter.

Event (N1)	Feeder	Start	Days	Detected by	First flag	Days flagged	Balance alarm
`partial_reduction_bypass`	F2	04 Feb	60	–	day 8	18%	day 13
`selective_bypass`	F2	09 Feb	65	–	day 3	5%	day 8
`meter_zeroing`	F1	11 May	20	IF, SVM	day 0	22%	–
`fdi_scale`	F3	02 May	63	R, z, SVM, AE	day 1	98%	day 14
`fdi_evasion`	F2	25 Apr	35	R, z, IF, SVM, AE	day 0	100%	day 9
`fdi_inflate`	F3	12 Sep	33	R, z, SVM, AE	day 1	97%	day 13
`fdi_replay`	F2	09 Aug	29	R, z, SVM, AE	day 0	100%	–
`coordinated_fdi`	F1	14 Oct	26	R, z, SVM, AE	day 0	100%	day 6
`bias_step`	F1	05 Jun	58	–	day 2	5%	day 14
`gain_drift`	F2	19 Apr	66	–	–	0%	day 15
`gain_drift`	F2	18 May	65	–	day 42	2%	day 16
`clock_shift`	F1	23 Feb	15	–	–	0%	–
`voltage_offset_drift`	F1	10 Apr	64	R, z, C, SVM, AE	day 0	100%	day 3
`stuck_value`	F1	12 Feb	3	R, z, IF, SVM, AE	day 0	100%	–
`stuck_value`	F2	28 Jun	1	R, z, IF, SVM, AE	day 0	100%	–
`stuck_value`	F3	10 Nov	2	–	–	0%	–

What the results show.

- **The stack is usable as a first filter.** It flags 2.3% of clean meter-days. The physics and domain rules alone flag none, yet they catch every data-only injection, the voltage offset and two of the three stuck meters from the first or second day (the third froze for 4 hours across midnight, below the 3-hour-per-day limit on either side).
- **No detector wins everywhere.** The One-Class SVM and the autoencoder carry the learned part (about three quarters of genuine-event days, most meter-fault events). Robust z is strong on falsification days but blind to genuine events. Isolation Forest is weak at this false-positive rate: most anomalies here shift values *inside* the normal range rather than isolate them.
- **The hard cases are the physically consistent ones.** Partial and selective bypass, the constant bias and both gain drifts keep V, I and E consistent and are flagged on at most 18%

of their days, as is a clock shift. This matches the difficulty ranking we gave in the data card.

- **The feeder balance sees them, but only as a feeder.** Its CUSUM fires 8–16 days after the start of both bypasses, the bias and both drifts. Pointing to the right meter fails: a household's daily share of its feeder varies far more than a 7–14% bias, and the gap itself moves: about 0.6 points of day-to-day noise on clean days, plus a spring–summer drop when PV homes export. Phase 2 will model the expected gap from feeder load, PV export and temperature, run CUSUM on the residual, and attribute alarms with longer windows.
- **Limits.** Simulated data from one generator; most falsification and sensor types occur once in N1, so these are case results, not rates. Feeder-level references use clean feeder-days.

5.5.2 Output to the Explanation Layer (Goal 2)

For every flagged meter-day, the detection layer passes on: the stack decision and each detector's score; which rules fired, with the measured value and threshold; the neighbourhood support (share of the feeder's 10 meters flagged the same day); the feeder-balance alarm state; and the autoencoder's per-feature reconstruction error, which tells Goal 2 which channel looks wrong.

6 Goal 2: Confidence Scores and Explainability

6.1 Goal and Requirements

The challenge requires that every flagged event comes with a verdict that a human can understand and verify. The system must output a confidence score, provide a plain-language explanation, and maintain an auditable decision log that records what was detected, concluded and why. It should use interpretable approaches and avoid black-box scoring whose reasoning cannot be examined.

Anomalies in this project are equipment and measurement problems, suspected data manipulation (theft or false data injection) and genuine electrical events. The component therefore chooses among four verdicts, the same four that the escalation layer of Goal 3 routes: `NORMAL`, `GENUINE_EVENT` (the data is right and the grid did something), `METER_FAULT` (a faulty or corrupted meter) and `SUSPICIOUS_DATA` (readings that look manipulated). A fifth outcome, `NO_DATA`, records a communication loss, which is not scored. The prototype runs on node N1 of the EnergyMind dataset: one transformer, 30 smart meters on three feeders of 10, readings every 30 minutes. Table 9 maps each requirement to a design element of this section.

Table 9. Requirements of Challenge Goal 2 and where this design meets them.

Requirement	Design element	Artifact produced
Confidence score	Calibrated probability of each of the four verdicts, obtained by multinomial Platt scaling of a few inspectable inputs (Section 6.6)	Fields `class_probabilities`, `confidence`, `band`; calibration curve, Brier score
Plain-language explanation	Rules + SHAP → evidence → structured summary → LLM text (no retrieval), checked against the rule trace, the SHAP ranking and the logged evidence before it is shown (Section 6.7)	Verdict card; field `llm_diagnosis` with the result of the faithfulness check
Auditable decision log	One hash-chained record per flagged meter-day (Section 6.8)	`decision_log.jsonl`
Record what was detected, concluded and why	Three explicit blocks in every record: `detected`, `concluded`, `why`	Rule trace, SHAP evidence, neighborhood support
Interpretable approaches	Rule-backed verdicts; SHAP on the detector scores; a calibration model whose coefficients can be printed	No verdict whose reasoning cannot be inspected

6.2 Inputs and Outputs of this Component

The detection layer of Goal 1 works on *meter-days* (48 readings of one meter) and, for every flagged meter-day, passes on: the stack decision and each detector's score (rules, robust z, CUSUM, Isolation Forest, One-Class SVM, autoencoder); which rules fired, with the measured value and threshold; the neighborhood support (share of the other 9 meters of the feeder flagged the same day); the state of the feeder energy-balance alarm; and the autoencoder's per-feature reconstruction error. This component adds a few evidence quantities computed from the same day's readings (Section 6.4) and produces, for every flagged meter-day, a verdict with its sub-type, the four class probabilities and a confidence band, the supporting evidence, a plain-language explanation and one decision-log record. The class probabilities are the input of the Goal 3 escalation layer.

6.3 Design Principles

1. **The LLM only describes.** The language model never detects an anomaly and never changes a verdict or a confidence value. This follows the conclusion of the rule-aware prompting work that the main value of an LLM here is a readable rationale for the operator rather than a replacement for the detector [17].
2. **Every statement is traceable.** Each statement in an explanation points to a logged item: a rule that fired, a feature value, or a SHAP contribution.
3. **Confidence is measured.** The score is calibrated on labeled events and its quality is evaluated like any other output of the system (Section 6.9).
4. **No explanation without evidence.** If the generated text fails the faithfulness check, the structured summary is shown instead.

6.4 Evidence Available to the Explanation Layer

We index a flagged meter-day by meter i and day t. Goal 1 supplies the detector scores, the fired rules and the neighborhood support $s_{i,t}$; the neighborhood-support idea follows prior work that adjusts an anomaly score using temporal context and the neighborhood [23]. This component computes the evidence vector $\mathbf{e}_{i,t}$ of Table 10 from the meter's 48 readings. We write E for the energy imported in a 30-minute interval ($\Delta t = 0.5$ h), V, I and PF for voltage, current and power factor, V^{head} for the voltage measured at the head of the feeder and X for the exported PV energy. Baselines (profiles, medians, scales) are learned on clean meter-days only, i.e. days whose 48 readings are all labelled normal.

Table 10. Evidence vector $\mathbf{e}_{i,t}$ of a flagged meter-day.

Symbol	Definition	What it tells a human
$v^0_{i,t}$	Share of readings with $V = 0$, or a `power_fail` entry in the meter's event log	Supply was lost: the meter is reporting a real outage
$g_{i,t}$	Share of missing readings	Communication loss
$\phi_{i,t}$	Longest run of readings in which energy *and* voltage are both unchanged (only while $V > 0$ and $X = 0$)	A stuck meter freezes every channel; real voltage never repeats for hours
$\rho_{i,t}$	Median over the day of $E \,/\, (V\,I\,\mathrm{PF}\,\Delta t/1000)$, using only readings with $V > 0$, $X = 0$ and an expected energy above 0.02 kWh	Electrical consistency: 1 when energy, voltage and current agree; skipped while the home exports PV
$s_{i,t}$	Share of the other 9 meters of the feeder flagged the same day (from Goal 1)	One house, part of the feeder, or the whole feeder
$\Delta V_{i,t}$	24-h median of $V - V^{\mathrm{head}}$	A meter cannot read more than its feeder head: a positive excess is a voltage-sensor offset
$\delta_{i,t}$	Relative drop of the day's energy below the meter's own 28-day profile	How much less than usual this home reports
$b_{i,t}$	Feeder energy-balance alarm (from Goal 1): on when the feeder's daily loss has moved away from its seasonal reference	Energy is disappearing (or appearing) somewhere on this feeder
$\kappa^{\mathrm{tamp}}_{i,t}$	`magnetic_tamper` or `cover_open` in the meter's event log	Weak hint of physical tampering (the log also has false alarms)
$c^{\mathrm{drift}}_{i,t}$	CUSUM alarm of Goal 1 on the meter's voltage offset, or slope of the meter's share of the feeder over 14 days	Slow drift or a persistent offset
$\pi_{i,t}$	Number of consecutive flagged days for this meter while its neighbors are not flagged	How long the deviation has lasted

Why these quantities, on this dataset. Three facts of node N1 shaped the list. (i) A supply loss is recorded as zeros, not as missing data: in N1 the 683 supply-loss readings of outages, load shedding, cable faults and the transformer trip have $V = 0$, while only communication losses are blank. Without v^0 a flat zero would look like a frozen meter. (ii) The consistency ρ is meaningful because the dataset's *data-only* injections edit the energy field alone ($\rho \approx 0.65$–0.69, or 1.68 for inflated readings), while honest meters already carry about 1% measurement error. Without the export gate, ρ would fire on 39% of the clean readings of N1's three PV homes; with it, on 0.25% of all clean readings. (iii) Sensor bias and physical bypass scale energy *and* current together, so ρ stays at 1 for them; what reveals a bypass is a large drop δ against the meter's own history together with the feeder balance b.

6.5 Rule-Backed Verdict Layer

The rule-backed scoring turns the evidence vector into a verdict and a sub-type (Figure 10). Rules are evaluated in a fixed order, so the first matching rule is the one reported, and its identifier appears in the log and in the explanation. The order follows what makes the other evidence

unreliable: a lost supply first (otherwise a flat zero looks like a frozen meter), then missing and frozen data, then physical consistency, which separates a manipulated or corrupted value from a real change, and only then the neighborhood support, which decides between a feeder-wide event and a local one. The split between sudden problems (R2, R3) and slow ones (R5, R5b) mirrors the split between point and contextual anomalies reported for smart-meter data [22].

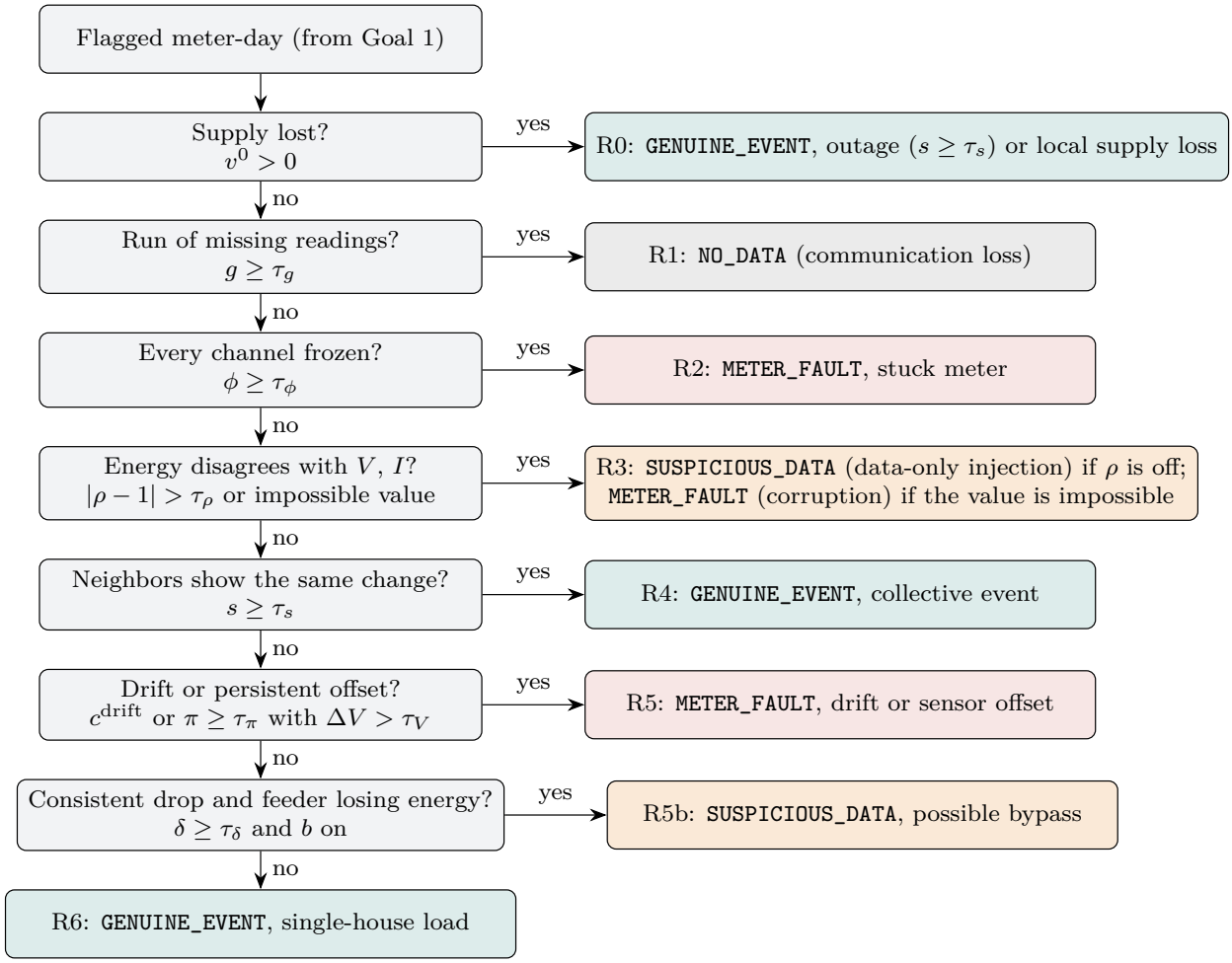


Figure 10. Rule cascade turning the evidence vector into a verdict. All thresholds τ are fitted by event-grouped cross-validation (Section 6.6) and stored in a versioned configuration file (`configs/thresholds.yaml`), not hard-coded.

Table 11. Rule table. The tamper flag κ^{tamp} is reported as supporting evidence for R5b but never decides a verdict: in N1 it appears on 4 falsified readings and on 55 clean ones.

Rule	Condition	Verdict	Reason sentence (template)
R0	$v^0 > 0$	`GENUINE_EVENT` (outage if $s \geq \tau_s$; local supply loss if 2–6 meters; single meter otherwise)	"Voltage was 0 V on {n} of {N} meters of feeder {f} for {hours} h: supply was lost and the meter is reporting correctly."
R1	$g \geq \tau_g$	`NO_DATA` (communication loss)	"{share} of this meter's readings did not arrive; no verdict is possible for this period."
R2	$\phi \geq \tau_\phi$	`METER_FAULT` (stuck meter)	"Energy and voltage have not changed for {hours} h; real voltage always varies."
R3	$\lvert\rho - 1\rvert > \tau_\rho$	`SUSPICIOUS_DATA` (data-only injection); `METER_FAULT` (corruption) if $E < 0$, $E = 99.999$ or $\times 1000$	"Reported energy is {ratio}× what voltage and current imply (limit {tau}); voltage and current look normal."
R4	$s \geq \tau_s$, ρ consistent	`GENUINE_EVENT` (collective)	"{n} of {N} meters on feeder {f} changed together and their electrical readings agree."
R5	$s < \tau_s$, c^{drift} on, or $\Delta V > \tau_V$ for $\geq \tau_\pi$ days	`METER_FAULT` (drift or sensor offset)	"This meter reads {dv} V above its feeder head, which is physically impossible, for {days} days."
R5b	$s < \tau_s$, ρ consistent, $\delta \geq \tau_\delta$, b on	`SUSPICIOUS_DATA` (possible bypass)	"This home reports {pct}% less than its usual level for {days} days while feeder {f} loses {gap}% more energy than usual; voltage and current agree with the reported energy, which fits a physical bypass."
R6	$s < \tau_s$, ρ consistent, short-lived	`GENUINE_EVENT` (single-house load)	"Only this home changed, briefly, and energy, voltage and current agree: an appliance switching on is the likely cause."

Notes on the rules.

- **R0 before R2.** Without it, every outage, load-shedding period, cable fault and transformer trip of N1 would be reported as a stuck meter, the opposite of the right answer.
- **R3 before R4.** A coordinated injection hits three meters of one feeder, so it could pass the neighborhood test; its energy disagrees with V and I, so R3 catches it first.
- **Suspect channel.** For R3 the suspect channel is energy by construction. For the other verdicts it is the feature with the largest autoencoder reconstruction error passed on by Goal 1; for faults that scale every channel (bias, bypass, frozen meter, clock shift) no single channel is named.
- **R5b before R6.** An isolated, electrically consistent drop is not automatically an appliance: when it lasts and the feeder starts losing energy at the same time, a bypass is the better explanation. A short, isolated, consistent change with no balance alarm stays a single-house load.
- **What the rules cannot settle.** Goal 1 found that small sensor biases and gain drifts are rarely flagged at all, and that the balance alarm names a feeder, not a meter. When b is on but no meter shows a clear drop, the case is logged as `SUSPICIOUS_DATA` with a low confidence

band, which sends it to a human (Goal 3).

6.6 Confidence Score

For each flagged meter-day the component outputs a probability for each of the four verdicts, $p_{i,t}(c)$ for $c \in \{\texttt{NORMAL}, \texttt{GENUINE_EVENT}, \texttt{METER_FAULT}, \texttt{SUSPICIOUS_DATA}\}$. The confidence of the reported verdict is $c_{i,t} = p_{i,t}(\hat{c})$, and the full vector is passed to Goal 3, which needs the margin between the two most likely verdicts and the set of plausible alternatives. The probabilities are built from three inputs, each of which can be inspected on its own:

1. the Goal 1 stack score $\tilde{a}_{i,t}$: the largest detector score, each standardized against its distribution on clean meter-days;
2. the *rule margin* $m_{i,t} \in [0, 1]$, which measures how clearly the rule that fired was satisfied. Each condition of that rule compares an evidence value e_j with a threshold τ_j. For each condition we compute its relative distance from the threshold, $\min\{1, |e_j - \tau_j|/\tau_j\}$, which is 0 when the value sits exactly on the threshold and 1 when it is at least one threshold-width away. The margin $m_{i,t}$ is the smallest of these values over the conditions of the rule, so a verdict is only as clear as its most borderline condition;
3. the *detector agreement* $\kappa_{i,t} \in \{0, \ldots, 6\}$: how many of the six Goal 1 detectors flagged the day, in the spirit of the explicit sigma rule used in rule-aware prompting [17].

Together with a one-hot code of the rule that fired, these inputs are mapped to probabilities by multinomial Platt scaling, one of the calibration techniques listed in the trust-calibration literature [19]: a multinomial logistic regression whose coefficients can be printed and read,

$$p_{i,t}(c) \;=\; \frac{\exp\big(\beta_{c,0} + \beta_{c,1}\tilde{a}_{i,t} + \beta_{c,2}m_{i,t} + \beta_{c,3}\kappa_{i,t} + \gamma_{c,r(i,t)}\big)}{\sum_{c'} \exp\big(\beta_{c',0} + \beta_{c',1}\tilde{a}_{i,t} + \beta_{c',2}m_{i,t} + \beta_{c',3}\kappa_{i,t} + \gamma_{c',r(i,t)}\big)}, \tag{5}$$

where $r(i, t)$ is the rule that fired. The target is the true class of the meter-day, known from the dataset labels. If the calibration curve is poor, isotonic regression, another technique from the same literature [19], is the alternative.

Confidence band. The probability is shown to operators as a band: HIGH above 0.90, LOW below 0.60, MEDIUM in between (Goal 3 thresholds). A class's band may be HIGH only if its calibration has been validated on at least 30 out-of-fold events. In N1 this holds for `GENUINE_EVENT` (37 events) and `METER_FAULT` (33), not for `SUSPICIOUS_DATA` (8 events, one per manipulation type), so `SUSPICIOUS_DATA` verdicts are capped at MEDIUM and are always reviewed by a human.

We do not assume that the result is calibrated; we measure it (Section 6.9). This matters because the confidence-based routing of Goal 3 assumes reasonably calibrated confidence: an overconfident model may delegate dangerous decisions to the AI, and an underconfident one may defer everything to humans [19].

Data used for fitting. A split by calendar period does not work on this dataset: in N1, November–December contain no falsification and no sensor-bias event, and September–October only two falsifications. Baselines (profiles, medians, scales) therefore use clean meter-days only, and the thresholds τ and the coefficients β, γ are fitted by 5-fold cross-validation over the whole year, with folds grouped by event (`gt_event_id`), so that no event is split between fitting and testing; within each training fold, τ is fitted on one half of the events and β, γ on the other. Clean days are assigned to folds in meter-week blocks to keep near-duplicate neighboring readings from leaking between sets. All reported numbers are out-of-fold, per event, with Wilson 95% intervals.

6.7 From Evidence to Plain Language

Figure 11 shows how the verdict and the evidence become the text that the operator reads, following the chain rules and SHAP → evidence → structured summary → LLM text. There is no retrieval

step (see "Why no retrieval" below). Full explanations are generated only for flagged meter-days, never for every normal reading, to keep latency and cost down.

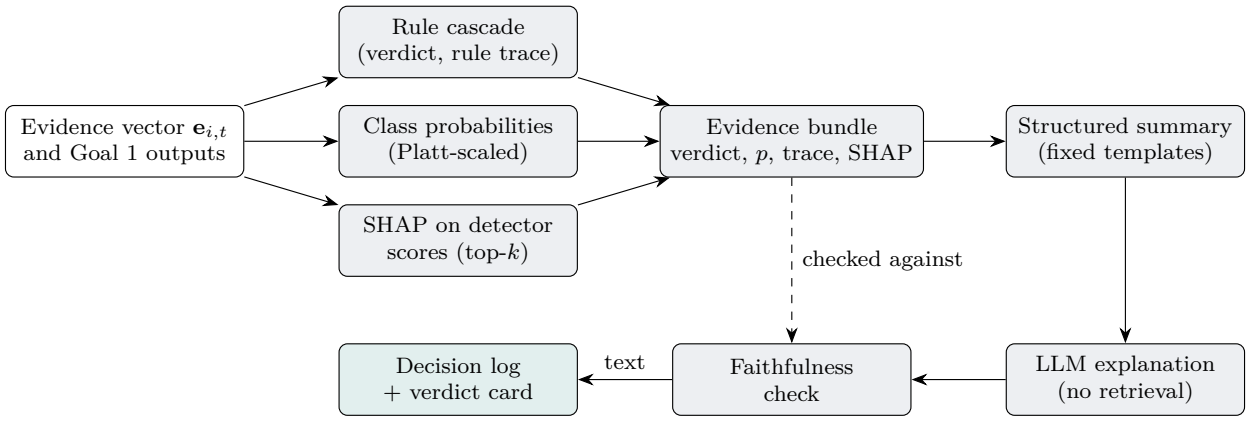


Figure 11. Explanation pipeline. The language model only rewrites the structured summary; it receives no retrieved documents, and its text is shown only if it passes the faithfulness check.

The language model receives only the evidence bundle and the structured summary, and writes about them; the faithfulness check compares its text with the same bundle. If the check rejects the text, the structured summary is logged and shown instead.

Step 1: attributions with SHAP

The rules say what kind of problem it is; SHAP says why the detectors flagged the meter-day in the first place. SHAP assigns each input feature a value showing how much it pushed a detector's score for this day up or down relative to a baseline, and it is model-agnostic. We apply the `shap` library to the Isolation Forest and the autoencoder of Goal 1, on their daily features, and only on flagged meter-days. Each feature is tied to a sentence template by a small lexicon, which is how a number becomes a statement (Table 12).

Table 12. Excerpt of the feature lexicon that turns attributions into sentences (one entry per Goal 1 feature; the evidence quantities of Table 10 have their own sentences in Table 11).

Goal 1 feature	Sentence template
Profile residual	"Today this home used {x} kWh against a usual {y} kWh for this season and day type."
Share of the feeder	"This home's share of feeder {f} has moved by {pct}% compared with the last four weeks."
Physics ratio	"Reported energy is {ratio}× what voltage and current imply."
V minus feeder-head V	"This meter reads {dv} V {above/below} the head of its feeder; usually {dv0} V."
Longest frozen run	"Energy and voltage stayed identical for {hours} h."
Lag gain against profile	"Today's pattern matches the usual one best when shifted by {h} h (clock problem?)."
Similarity to last week	"Today's readings are almost identical to the same day last week (possible replay)."

Step 2: structured evidence summary

A deterministic step assembles the evidence bundle into a summary: a headline (verdict, channel, confidence and band), the strongest reasons (the rule's reason sentence first, then the highest

SHAP contributions), the runner-up verdict, a *what would change this verdict* line generated from the rule thresholds, and a suggested check. This follows the recommendation, in a review of AI approaches for smart-grid reliability, to combine local feature attribution, counterfactuals and logged explanations [21]. Every sentence comes from a template of Table 11 or Table 12, filled with values from the log. The summary is the input of the language model and also the text shown if the faithfulness check fails. An illustrative card (modelled on the data-only injection of feeder N1-F3 in May; values for illustration):

```
Event N1-2025-05-03-M026 | Meter N1-M026 (feeder N1-F3) | 2025-05-03
VERDICT: SUSPICIOUS DATA, energy channel     CONFIDENCE: 0.74 (MEDIUM)
Runner-up: METER FAULT (0.19)

Why:
 1. Reported energy is 0.65x what voltage and current imply
    (limit +/-5%) -> rule R3.
 2. Isolated: 0 of 9 other meters on feeder N1-F3 changed.
 3. Voltage and current follow this home's usual pattern; only the
    energy field is low.

This would be a METER FAULT instead if: the value were impossible
(negative, x1000, 99.999); a GENUINE EVENT if >= 3 neighbors changed
the same way and energy matched voltage and current.

Suggested check: read the meter register on site and compare it with
the reported energy; inspect the data path (gateway, head-end).
```

Step 3: LLM explanation with a faithfulness check

The language model turns the structured summary into a readable diagnosis in plain language. It receives the evidence bundle as structured data and nothing else. It is asked to justify the verdict against an explicit, stated rule, as in rule-aware prompting for grid telemetry [17], and it may not alter the verdict, the confidence or the band.

Before the text is displayed, a faithfulness check compares it with the evidence bundle:

- every reason it names must be a fired rule or one of the top-k SHAP features;
- every number it quotes must match the logged value up to rounding;
- the verdict, the confidence and the band must be those in the log.

If any test fails, the structured summary is shown instead and the event is logged with that fact. We treat the LLM as a layer that describes the evidence, not as a source of reasoning.

Why no retrieval (RAG). We considered giving the language model retrieved context (descriptions of the fault kinds, the anomaly taxonomy, earlier log entries for the same meter). We dropped it. A study of retrieval-augmented LLM post-processing for streaming anomaly detection found only a modest gain, and the faithfulness of the explanations did not improve with retrieval [18]. For an operator who must verify a verdict, that small gain does not justify the extra latency on every flagged case and a document corpus to build and maintain. Everything the explanation needs is already in the evidence bundle.

6.8 Auditable Decision Log

Every flagged meter-day produces one record in an append-only JSON Lines file. Each record stores the hash of the previous one, so a later edit or deletion breaks the chain. The record is grouped around the three questions the challenge asks (Table 13); an abridged example follows. The escalation layer of Goal 3 appends its own `governance` block (route, rules fired, operator judgement, final decision) to the same record; this section defines only the fields produced by

detection, conclusion and explanation. The full schema is `docs/decision_log_schema.json` in the project repository.

Table 13. Structure of a decision log record.

Block	Fields
`detected`	event id, day, node, feeder and meter id, model version, stack score, per-detector scores, per-feature reconstruction error, flag
`concluded`	verdict, sub-type, suspect channel, class probabilities, confidence, band
`why`	rule trace (rule id, measured value, threshold, passed or not), SHAP evidence, neighborhood support, feeder-balance alarm, threshold version
`explanation`	LLM diagnosis, result of the faithfulness check, text actually shown
`ground_truth`	filled afterwards, for evaluation only: dataset label, sub-type, event id


```
{"event_id":"N1-2025-05-03-M026","ts":"2025-05-03","node_id":"N1",
 "feeder_id":"N1-F3","meter_id":"N1-M026",
 "detected":{"model_version":"detect-0.1","fused_score":4.8,"flag":true,
   "detector_scores":{"rules":1,"robust_z":9.1,"svm":2.3,"ae":6.4}},
 "concluded":{"verdict":"SUSPICIOUS_DATA","subtype":"data_only_injection",
   "suspect_channel":"energy","confidence":0.74,"band":"MEDIUM",
   "class_probabilities":{"NORMAL":0.02,"GENUINE_EVENT":0.05,
                          "METER_FAULT":0.19,"SUSPICIOUS_DATA":0.74}},
 "why":{"rules":[{"id":"R3","measured":0.65,"threshold":0.05,"passed":true},
                 {"id":"R4","measured":0.0,"threshold":0.3,"passed":false}],
        "shap":[["physics_ratio",0.52],["profile_residual",0.21]],
        "neighbour_support":"0/9","balance_alarm":false,
        "thresholds_version":"0.1-phase1"},
 "explanation":{"llm_diagnosis":"...","faithfulness":"pass","shown":"llm"},
 "prev_hash":"9f2c...","hash":"41ab..."}
```


(The values are invented for illustration.) The record keeps the exact inputs and the version of the thresholds and models, so a verdict can be reproduced and reviewed later.

6.9 How We Will Evaluate Explanations and Confidence

The dataset labels every reading with its class, its exact sub-type and the event it belongs to, so verdicts and explanations can be tested against ground truth rather than only judged by eye. In N1 the flagged material comes from 98 events: 37 genuine events, 33 meter faults (25 of them data corruption), 8 falsifications and 20 communication losses. Most manipulation and sensor-fault types occur once, so results are reported per event, with intervals, and never as a single accuracy figure; we use precision, recall and false-positive rate per class.

Table 14. Evaluation plan for confidence scores and explanations.

Aspect	Measure	Method
Confidence quality	Calibration curve and Brier score [20]; verdict accuracy within each band	Out-of-fold, per class and overall
Verdict correctness	Precision, recall and false-positive rate per class, with event counts	Against the dataset labels, out-of-fold; also on the *subtle set*
Explanation correctness	Share of verdicts whose named channel is the true one (Table 15); share whose scope (isolated, part of the feeder, whole feeder) is right	Labels versus the `why` block
LLM faithfulness	Share of LLM texts accepted by the faithfulness check; kinds of statements rejected	Automatic check on all flagged meter-days
Does the LLM help	LLM text versus the plain structured summary, rated on clarity and correctness (1–5) by several team members who do not know which is which	Sample of flagged meter-days
Response latency	Median and 95th-percentile time of rules, SHAP and LLM per flagged meter-day	Timer on the evaluation run

Table 15. The dataset has no "injected channel" column; this fixed mapping from sub-type to channel is used to score the suspect channel named in the explanation.

True channel	Dataset sub-types
Energy	`fdi_scale`, `fdi_evasion`, `fdi_inflate`, `fdi_replay`, `coordinated_fdi`, all data corruption (`spike_gain`, `negative_value`, `saturation`, `unit_error_x1000`)
Voltage	`voltage_offset_drift`
All channels	`bias_step`, `gain_drift`, `stuck_value`, `clock_shift`, `partial_reduction_bypass`, `selective_bypass`, `meter_zeroing`: only the verdict and the scope are scored

The *subtle set* is built from existing events, without changing the dataset: the first seven days of the two gain drifts and of the bias step, and the evasion attack, whose scaling factor wanders to mimic natural variation.

6.10 Risks and Limitations

- **Few events per type.** Most manipulation and sensor-fault types occur once in N1, so per-type results are case studies, not rates; the `SUSPICIOUS_DATA` band is capped at MEDIUM for that reason.
- **Simulated data.** Thresholds and calibration coefficients are fitted on one generator's anomalies and may be optimistic; results on the subtle set are reported alongside.
- **Physically consistent manipulation.** Bypasses keep energy, voltage and current consistent; R5b depends on the feeder-balance alarm of Goal 1, which names a feeder and reacts after 1–2 weeks. A meter that under-reports slightly may never reach R5b.
- **PV homes.** The consistency check is skipped while a home exports, so an injection limited to export hours would be missed by R3.

- **Weak tamper log.** Tamper codes also appear on clean readings; they support but never decide a verdict.
- **Correlated features.** SHAP can spread credit among correlated features. The rule trace is therefore the primary explanation and SHAP the supporting one.
- **Cost of SHAP.** SHAP is run only on flagged meter-days and its latency is measured; the rule trace remains available if it is too slow.
- **Calibration can drift** across seasons or meters; the calibration curve is checked again on later events.
- **Convincing but wrong.** A fluent explanation can raise trust beyond what the evidence justifies, so every card shows the confidence band, the runner-up verdict and the counterfactual line.
- **Language model errors.** The faithfulness check and the fallback to the structured summary limit this, and the acceptance rate of the check is reported.

7 Goal 3: Human–AI Escalation and Governance

The detection layer (Goal 1) answers "is something abnormal?" and the diagnosis layer (Goal 2) answers "what is the most likely explanation?". The governance layer answers a different question: *who should decide or act, and under which conditions?* That cannot be reduced to model confidence. An anomaly score says how unusual the data look, not why; a correct diagnosis says nothing about the cost of being wrong; and the right level of autonomy depends on the action — requesting another reading is harmless, switching a feeder or declaring a customer fraudulent is not. The prototype runs on node N1 and controls no real STEG equipment; all thresholds are prototype parameters, not STEG values. The complete design is in `docs/goal3_governance`.

7.1 Principles and research foundations

Table 16. Design principles of the governance layer. The goal is appropriate reliance, not full automation.

#	Principle	Consequence in the design
P1	Autonomy is granted per action, not per event	One event may trigger safe automatic actions while its consequential action stays locked behind approval
P2	Escalation is monotone	Rules may raise the governance level, never lower it; high confidence cannot override critical risk
P3	Independent human judgement first	In reviewed cases the operator records a diagnosis and confidence before seeing the AI output
P4	No default privilege	Neither AI nor operator wins a disagreement by default; strong disagreement triggers evidence gathering and a second review
P5	Optimise the human + AI system	Success is measured on the combined system (accuracy, safety, workload), not on AI accuracy alone

Human-aligned calibration [24]. When a decision maker has her own confidence H and sees the AI's confidence B, calibration of B alone is not enough: there are processes in which B is perfectly calibrated yet every monotone (rational) decision policy is suboptimal, because calibration can fail within a given human confidence level. Near-optimal monotone policies exist when the positive rate increases jointly in both confidences ("alignment"), which multicalibration per human level guarantees. Empirically, people move toward the AI's stated confidence whatever their prior. *We take from it*: treat operator and AI confidence jointly; elicit operator confidence on three levels *before* showing the AI; log both; audit alignment.

Learning to defer [25]. A classifier h and a rejector r are trained on the loss of the combined system. The Bayes-optimal rule defers exactly when the expert is at least as likely to be right:

$$h^B(x) = \arg\max_y \eta_y(x), \qquad r^B(x) = \mathbb{1}\big[\max_y \eta_y(x) \le P(M = Y \mid X = x)\big], \tag{6}$$

and the share of cases handled alone, the *coverage*, is the central trade-off. *We take from it*: deferral as an explicit outcome, coverage as a primary metric, the coverage sweep, the (x, y, m) logging requirement and synthetic experts for prototyping. Neither paper treats safety constraints, irreversible actions or asset criticality; those are EnergyMind design decisions below.

7.2 Conceptual design

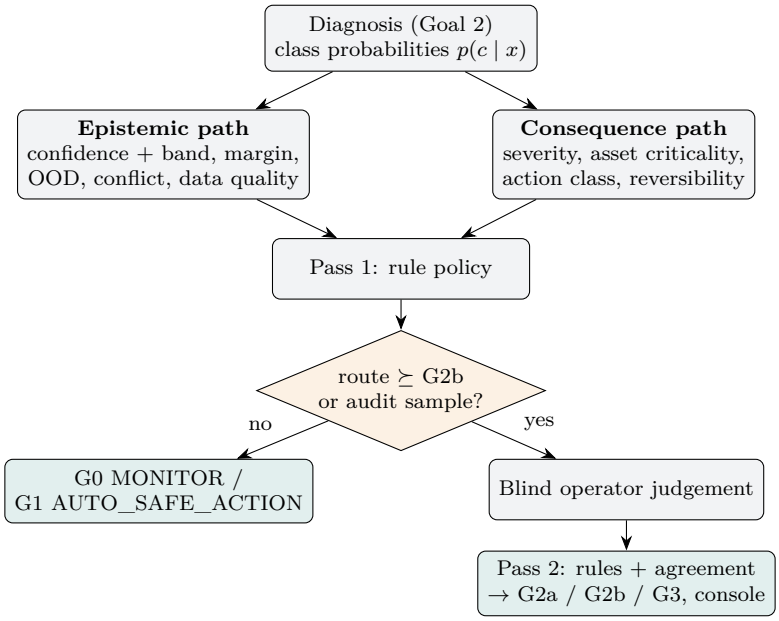


Figure 12. Runtime decision flow. The epistemic and consequence paths stay separate until the policy is applied, so confidence cannot dilute risk; the blind judgement comes before final routing because agreement is an input.

Inputs. From Goal 2, per flagged meter-day: the calibrated probability of each of `NORMAL`, `GENUINE_EVENT`, `METER_FAULT` and `SUSPICIOUS_DATA`, the band and the rule that fired; from Goal 1, the anomaly score, the neighbourhood support and the feeder-balance alarm. `NO_DATA` (communication loss) is handled by rule R0. In N1 the classes cover 37 genuine events, 33 meter faults, 8 falsifications and 20 communication losses.

Five quantities that must not be merged. Anomaly score (how unusual), AI confidence (how sure), severity (how significant technically), asset criticality (how important the asset) and risk (what happens if the governance decision is wrong). Only risk is derived from the others, by an explicit rule.

Table 17. Asset criticality and severity. The dataset has no critical asset and no criticality or severity field; both are a documented synthetic overlay, joined on identifiers without modifying the data.

Asset (registry `configs/assets.yaml`)	**Criticality**	**In N1**
Residential meter	LOW	24 meters
Small commercial meter	MEDIUM	6 meters (F1: 2, F2: 3, F3: 1)
Feeder-head meter, feeder switching	HIGH	3 feeders
Feeder N1-F3 (assumed to supply a health centre)	CRITICAL	10 meters
Scope and duration	**Severity**	
One meter, less than 6 hours	LOW	
One meter for 6 hours or more, or 2–6 meters	MEDIUM	
A whole feeder (10 meters)	HIGH	
The whole node or the transformer	CRITICAL	

Table 18. Governance levels. G0–G1 are "AI handles"; G2a–G3 are "defer". Above G1, only G1 actions may run automatically; locked actions never run on timeout.

Level	Route	Who decides	Typical case
G0	MONITOR	AI	`NORMAL`, confident, familiar
G1	AUTO_SAFE_ ACTION	AI (whitelisted reversible, non-physical actions)	confident freeze of a residential meter
G2a	AI_ RECOMMENDATION	AI proposes, operator acknowledges	probable meter fault, inspection suggested
G2b	HUMAN_REVIEW	operator, after blind judgement	uncertainty, disagreement, OOD, suspicious data, high risk
G3	HUMAN_ APPROVAL_ REQUIRED	named operator with written justification (two people if risk is CRITICAL)	switching, disconnection, fraud declaration, critical asset

7.3 Risk and confidence

Confidence bands. With $\hat{c} = \arg\max_c p(c \mid x)$ and $p^\star = p(\hat{c} \mid x)$, the band is HIGH if $p^\star \geq \tau^{\text{high}}_{\hat{c}}$, LOW if $p^\star < \tau^{\text{low}}_{\hat{c}}$, MEDIUM otherwise ($\tau^{\text{high}} = 0.90$, 0.85 for `GENUINE_EVENT`; $\tau^{\text{low}} = 0.60$; minimum margin $p_{(1)} - p_{(2)} = 0.30$). The band is capped at MEDIUM when the input is out of distribution, data quality is not good, or the class's calibration is not validated — which requires at least 30 out-of-fold events. In N1 this holds for `GENUINE_EVENT` (37) and `METER_FAULT` (33) but not `SUSPICIOUS_DATA` (8), so every suspicious-data case is reviewed by a human.

Risk. The wrong-action risk of an action a on an asset of criticality κ is $\rho_{\text{act}}(a) = \max(\beta(a) \oplus \mathbb{1}_{\text{irrev}}(a), \kappa)$ for physical or customer-affecting actions and $\beta(a) \oplus \mathbb{1}_{\text{irrev}}(a)$ otherwise, with $\beta(a)$ the base risk of the action class and $\oplus 1$ one level up. The missed-event risk uses the most severe *plausible* hypothesis, so a confident `METER_FAULT` cannot hide a plausible `GENUINE_EVENT`:

$$\sigma^\star(x) = \max_{c:\, p(c|x) \geq 0.05} \sigma(c), \qquad \rho_{\text{miss}} = M[\kappa, \sigma^\star], \qquad \rho_{\text{case}} = \max\big(\rho_{\text{miss}}, \max_a \rho_{\text{act}}(a)\big), \tag{7}$$

where M is an ordinal criticality $\times$ severity matrix (e.g. LOW criticality with HIGH severity gives MEDIUM risk; CRITICAL criticality with HIGH severity gives CRITICAL). A maximum is used, never a weighted sum: ordinal ranks cannot be averaged, and "low severity" must never cancel "health-centre feeder". The price is over-escalation, which is measured rather than assumed acceptable.

7.4 Escalation policy

Order the routes G0 $\prec$ G1 $\prec$ G2a $\prec$ G2b $\prec$ G3; each rule sets a floor and $\text{route}(e) = \max_k f_k(e)$.

Table 19. Routing rules. R6 follows the dataset's reference responses: a drift or bias that bills a customer wrongly for weeks needs a human; scheduled load shedding and short sags only need logging.

Rule	Condition	Floor / effect	Tunable
R0 Input validity	missing inputs (`NO_DATA`), unregistered asset, unapproved model	$\geq$ G2b; short communication loss ($\leq$12 intervals): G1 gap filling	no
R1 Hard action	physical, customer-affecting, accusatory or irreversible action	action locked; G3	no
R2 Consequence	case risk CRITICAL, or CRITICAL asset with a non-`NORMAL` diagnosis	$\geq$ G2b, consequential actions locked, two-person approval if CRITICAL	no
R2b	case risk HIGH	$\geq$ G2b	no
R3 Agreement	strong / partial disagreement, weak agreement	$\geq$ G2b (+ second review, more data, freeze if strong)	no
R4 Familiarity	out of distribution, insufficient data	$\geq$ G2b + more data	thresholds
R5 Confidence	LOW band or strong conflict / MEDIUM band or small margin	$\geq$ G2b / $\geq$ G2a	yes
R6 Diagnosis default	`NORMAL` $\rightarrow$ G0. `METER_FAULT` $\rightarrow$ G1, $\geq$ G2a for a voltage offset or clock shift, $\geq$ G2b if it lasts 24 h or more or affects billing (gain drift, bias). `GENUINE_EVENT` $\rightarrow$ G0 if scheduled or self-clearing (load shedding, short sag, surge), G2a otherwise, $\geq$ G2b for an equipment cause of severity $\geq$ MEDIUM. `SUSPICIOUS_DATA` $\rightarrow$ G2b	floor	yes
R7 Recurrence	too many automatic actions on one asset	$\geq$ G2b	yes
R8 Workload	queue strained	priority only, never the route	yes
R9 Consistency	locked action $\rightarrow$ G3; proposed action $\rightarrow$ $\geq$ G2a	floor	no

Critical consequence dominates confidence. For any non-`NORMAL` diagnosis, every hard action is locked and routed to G3, and a CRITICAL asset is always at least G2b, *for every value of the AI confidence*: R1, the criticality branch of R2 and R9 do not depend on $p^\star$, and a maximum of floors cannot be lowered by the floors that do. Confidence may resolve doubt about *what happened*, never change *what is at stake*. The property is an automated test on 400 random cases (`tests/test_governance.py`).

Two passes and agreement. Pass 1 applies every rule except R3; if the route is G2b or above, or the case falls in the 5% audit sample, the operator gives a blind diagnosis and a LOW/MID/HIGH confidence. Pass 2 adds R3, which can only raise the route. Agreement states: AGREEMENT (same class, both at least MID; never lowers the route), WEAK_AGREEMENT (same class, someone guessing; $\geq$ G2b + more data), PARTIAL (competing hypotheses; $\geq$ G2b) and STRONG (different classes, the AI gives the operator's class less than 0.20, and both are confident or the pair is critical, e.g. `GENUINE_EVENT` vs. `METER_FAULT`; $\geq$ G2b + second review + more data + freeze). When neither party is confident, at most two rounds of safe data requests (neighbour readings, feeder-head voltage, history, feeder balance) are run before a supervisor takes the case; new evidence may raise the route, never lower it.

7.5 Operator interaction and audit

Against over-trust: blind-first judgement; confidence as a band with its real validation count, and no count at all where calibration is not validated (on N1: `SUSPICIOUS_DATA`); the runner-up always shown; equal buttons with no default; written justification for G3 approvals and

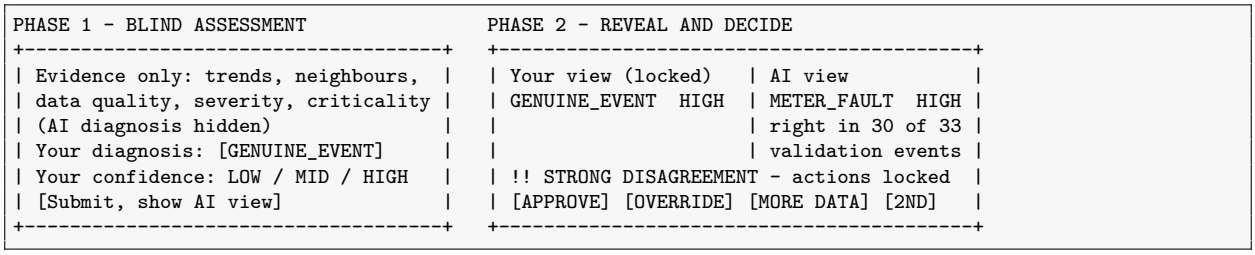


Figure 13. Operator console mock-up: blind assessment first, then the AI's view is revealed (counts illustrative).

overrides; two-person approval for CRITICAL risk. **Against under-trust:** routine events never reach the operator; track records of AI and operators per event type; outcome feedback; similar past incidents.

Every decision is appended to the hash-chained decision log shared with Goal 2, as a `governance` block (severity, criticality, per-action risk, blind and post-reveal judgements, agreement, route and fired rules, final decision and justification, policy version). The expert label used for learning is the *blind* judgement (the post-reveal one echoes the AI), and the operator's decision is never treated as ground truth. A calibration phase replays labelled dataset events to estimate each operator's accuracy per class, with Wilson intervals.

Implementation. The Version 1 engine (bands, risk, agreement, locks, routing R0–R9) is `src/energymind/govern/policy.py`; the six scenarios below and five safety invariants, each on 400 random cases, are automated tests. The request-more-data loop, the console and Version 2 are design.

Version 2. A learned rejector replaces only the tunable rules R5–R6, inside the immutable rules: $\text{route}_{V2}(e) = \max\big(\max_{k\in\{R0,R1,R2,R3,R9\}} f_k(e),\ \phi(r_\theta(x))\big)$ with $\phi =$ G1 (AI handles) or G2b (defer). It can only move cases inside the region the rules already allow to automate, and is promoted only after a shadow-mode comparison with Version 1.

7.6 Evaluation

The layer is evaluated as a human–AI system on N1's labelled event stream, with synthetic operator profiles and AI outputs taken out-of-fold from Goal 2. Main metrics: coverage $|A|/N$; review rate; false-escalation rate (deferred although the AI was right and the risk was low); missed-escalation rate (automated although the AI was wrong or the risk required review); unsafe automatic actions (must be exactly 0); system accuracy and a cost-weighted loss; over- and under-reliance; decision time and workload. Two analyses follow the references: a coverage sweep (system accuracy and missed escalations against coverage, one curve per operator profile) and an alignment audit (positive rate per AI-confidence bin and operator level; cells with at least 30 samples, which N1 reaches only for `GENUINE_EVENT` and `METER_FAULT`). The out-of-distribution rule is exercised on N1's hard-negative periods (Ramadan, August vacations) and on its clock shift, which Goal 1 does not recognise.

Agreement with the dataset's reference policy. Each labelled reading carries a suggested response, which we defined in our data card as a starting point; each route is compared with it, a route below the reference counting as a missed escalation and above it as a false escalation. Deviations caused by the asset overlay (a sag on the critical feeder is reviewed although the reference says "log") are listed separately.

Table 20. Reference responses and their nearest governance level.

Reference response	Route	N1 readings
`auto_log`	G0	1 100 grid-event and 50 equipment readings (load shedding, sag, surge, cold-load pickup)
`auto_correct`	G1	112 corruption, 151 stuck meter, 78 short communication losses
`notify_operator`	G2a	500 outage/swell, 3 711 voltage-offset/clock-shift, 351 long communication losses
`human_decision`	$\geq$ G2b	14 175 falsification, 8 922 gain drift/bias, 3 244 equipment failure

Table 21. Reference scenarios built on real N1 events, run as automated tests in `tests/test_governance.py` (AI outputs and operator answers are scripted test inputs). S2 and S6 show the central claim: high confidence, even with human agreement, never authorises a consequential or accusatory action.

Scenario (N1 event)	AI output (scripted)	Outcome	Reference
S1 Stuck meter, N1-F1, 12–14 Feb	`METER_FAULT` 0.95	automatic safe action (quarantine, re-read, ticket); review if it outlasts 24 h	auto_ correct
S2 Voltage sag on the critical feeder N1-F3, 27 Jun	`GENUINE_EVENT` 0.98	transfer to backup feeder locked: two-person approval	auto_log (overlay)
S3 Gain drift, N1-F2, from 19 Apr	`METER_FAULT` 0.92, operator agrees	human review (billing); agreement does not lower it	human_ decision
S4 Demand surge, N1-F2, 23 Jun	`METER_FAULT` 0.93 vs. operator `GENUINE_EVENT`	strong disagreement: second review, more data, freeze	auto_log
S5 Clock shift, N1-F1, 23 Feb	`SUSPICIOUS_DATA` 0.46, OOD	review with a request for more data; no consequential action	notify_ operator
S6 Partial bypass, N1-F2, 4 Feb	`SUSPICIOUS_DATA` 0.94, capped at MEDIUM	disconnection locked: two-person approval; operator orders an inspection	human_ decision

Limitations. No real operator data (thresholds, risk matrix and agreement rules are engineering choices); synthetic criticality; only 8 suspicious-data events in N1, so their calibration cannot be validated; conservative by design; the policy decides which cases humans see, so logged data reflect the policy (the audit sample mitigates this); blind-phase latency may be too slow for urgent cases; an attacker aware of the policy could try to stay inside the automated region. All results are simulated and must be reported as such.

8 Goal 4: Shared Learning Across the Noosphere

8.1 Why a single node is not enough

A neighborhood node sees only a few examples of each failure mode. In our simulated year, each falsification type occurs only 2–3 times *in the whole dataset*, that is at most once per node, and several types never occur in a given node at all (Table 22). The hardest cases for a lone detector (`selective_bypass`, `partial_reduction_bypass`, `meter_zeroing`, `gain_drift`, `clock_shift`, `negative_value`, `transformer_degradation`) are exactly the rare ones; the Goal 1 run on node N1 confirms that bypasses, biases and drifts are the cases a single node detects worst. A node that has never seen an attack cannot recognise it; a node that has will recognise it earlier if its neighbours warn it.

Table 22. Falsification events per node in the dataset. A 0 means the node never sees that attack in the year.

Falsification type	N1	N2	N3
`coordinated_fdi`	1	**0**	1
`fdi_evasion`	1	1	**0**
`fdi_inflate`	1	**0**	1
`fdi_replay`	1	**0**	1
`fdi_scale`	1	1	**0**
`meter_zeroing`	1	1	**0**
`partial_reduction_bypass`	1	1	1
`selective_bypass`	1	**0**	1

Goal 4 turns the local trust layer of Goals 1–3 into a *collective immune system*: nodes exchange what they *learned*, never what they *measured*.

The design rests on three principles:

P1. Share knowledge, not data. Raw readings, identifiers, explanations and logs never leave the node.

P2. Trust nothing shared by default. Every incoming artifact is treated like an unverified meter reading: it is validated, canary-tested locally, and adopted only with operator approval.

P3. Sharing can never trigger grid action. The escalation rules of Goal 3 stay in force whatever the shared model says; shared knowledge only changes *scores*, never *authority*.

8.2 Nodes and honest scope

The evaluation uses our simulated dataset (1,051,200 half-hourly rows, 60 meters, 7 feeders, seed 2025). Each node is one smart-metered neighborhood (Table 23); public Tunisian statistics were used only to calibrate the simulator, so no result here describes a real STEG grid. Node N1 is the Phase 1 single-node prototype on which Goals 1–3 run; N2 and N3 are the other participants of the sharing design. As the challenge allows, multi-node sharing is a documented design in Phase 1: the protocol below is specified, and the experiments that would measure its benefit are planned in Section 8.8.

Table 23. Simulated nodes. Heterogeneity (climate, meter mix, feeder length, customer class) is what makes sharing useful; a common simulator limits how large it is.

Node	Profile	Meters	Feeders	Role in the experiments
N1	Coastal suburb	30	3	Largest node; local-baseline reference
N2	Hot interior town	15	2	Strong summer load, different climate regime
N3	Mixed residential/commercial	15	2	Different customer classes and load shapes

8.3 Architecture of the Noosphere

Figure 14 shows the architecture. Each node runs the full local trust pipeline and exposes only three narrow, versioned *channels* to a coordinator. A fourth, out-of-band *alert* path carries urgent attack warnings.

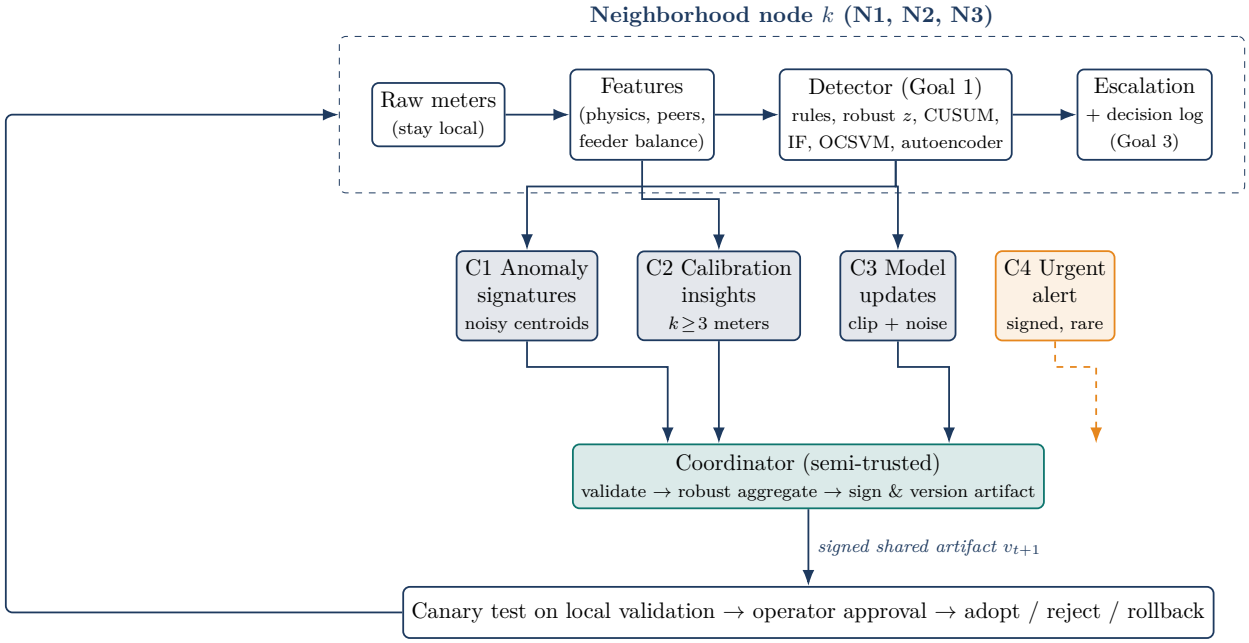


Figure 14. Noosphere architecture. Raw data and decisions stay inside the node; three protected channels and one urgent-alert path reach a coordinator, whose output is only adopted after a local canary test and human approval.

8.4 What is shared, what stays local, and why

Table 24. Sharing contract. Each row answers the challenge's three questions: what, what stays local, why.

Channel	Shared artifact	Safeguards	Why it helps
C1 Anomaly patterns	Per-node centroids of "trusted" and "untrusted" records in a robustly standardised feature space (Eq. 8); optional per-*family* centroids when $\geq n_{\min}$ events exist: physical bypass and data-only FDI (Goal 2 verdict `SUSPICIOUS_DATA`), sensor drift and corruption (`METER_FAULT`)	Feature clipping, Gaussian noise on released centroids, coordinate-wise median across nodes, count bucketing	Lets a node recognise an attack family it has seen too rarely to learn, without exchanging any time series
C2 Calibration insights	Median of $\log\big(E/(V\,I\,\mathrm{PF}\,\Delta t/1000)\big)$ per *meter model* on clean, non-exporting readings, with sample size bucket	Release a model only if ≥ 3 meters contribute; medians only; no meter IDs	Separates a model-wide calibration bias (firmware, batch) from a single faulty meter
C3 Model updates	Clipped, noised updates to a shared logistic-regression screening model	Update-norm clipping C, Gaussian noise σ, robust aggregation, secure aggregation (deployment)	Nodes with few rare events borrow strength from others; the shared model supports, never replaces, the local detectors of Goal 1
C4 Urgent alert	Signed "attack fingerprint": subtype family, feature-residual profile, affected meter-model, severity	Human-confirmed at the source node; rate-limited; expires; receivers treat it as a *prior*, not a verdict	Days-to-hours propagation of a confirmed coordinated attack

Stays local (never leaves the node): raw readings and timestamps; meter, customer and feeder identifiers; coordinates and dwelling data; per-meter reference values and the local median/MAD scaling statistics; per-record features, verdicts and SHAP explanations; operator notes and overrides; full decision logs; local detector models. *Reason:* half-hourly smart-meter traces reveal occupancy, sleep and appliance use [26, 27]; explanations and logs are accountability records that belong to the operator that made the decision.

8.5 Protocol

The protocol defines how independent neighborhood nodes prepare local information, share privacy-aware summaries or model updates, and evaluate the knowledge received from the coordinator. Raw meter readings and record-level decisions remain local. The protocol distinguishes four sharing channels: anomaly signatures (C1), calibration insights (C2), federated model updates (C3), and urgent alerts (C4). C1–C3 are regular knowledge-sharing mechanisms, whereas C4 is a separate, rare alert path.

8.5.1 Protocol overview

Figure 15 presents the complete protocol, from local data preparation to the local acceptance or rejection of a shared model. The multi-node communication and secure aggregation components

are documented design requirements; the figure describes the proposed architecture rather than claiming that the complete distributed system has been deployed.

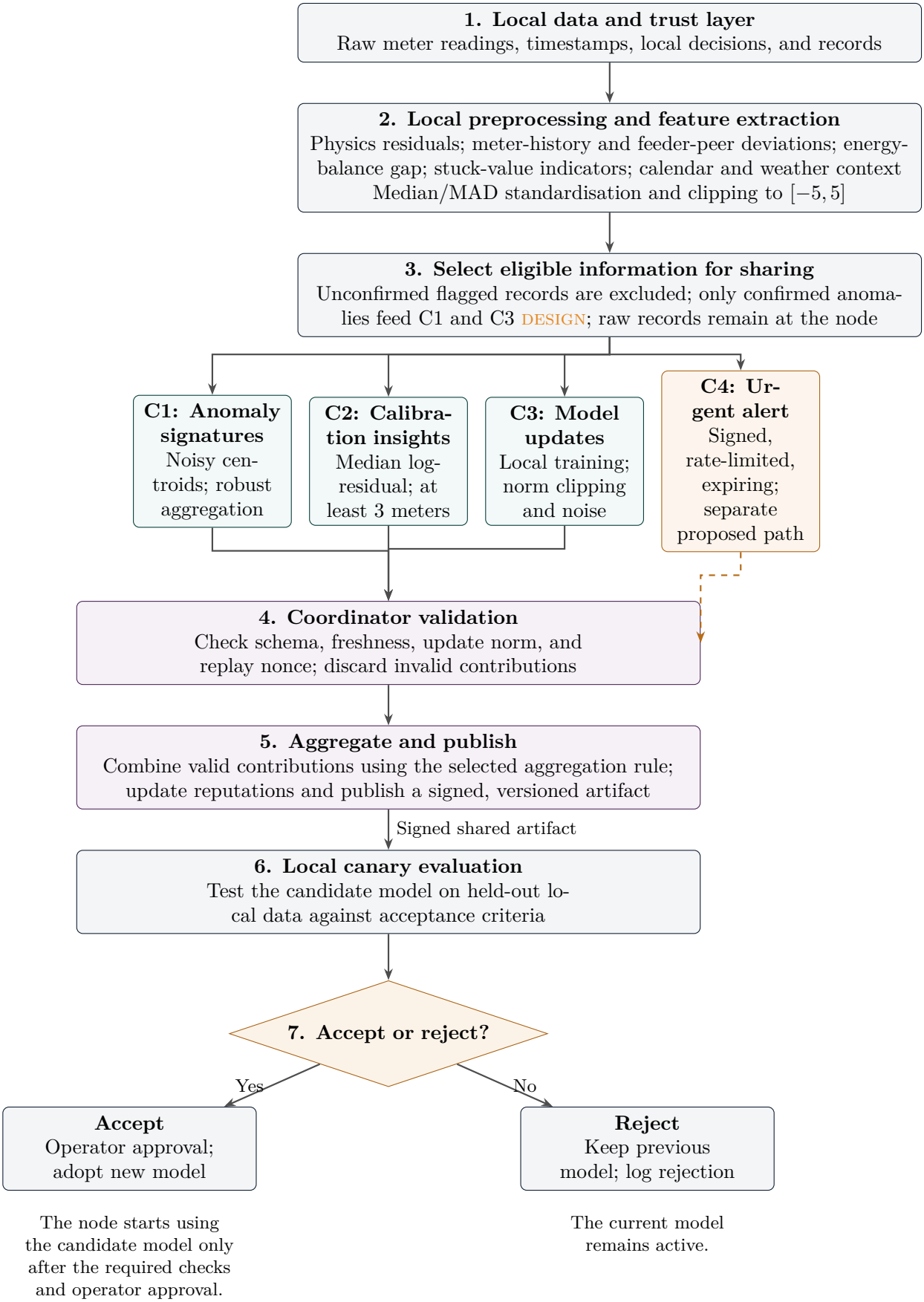


Figure 15. Proposed shared-learning protocol. Raw readings and record-level decisions remain local. C1–C3 carry regular shared information; C4 is a separate proposed urgent-alert path. A shared model is adopted only after local evaluation and operator approval.

8.5.2 Step 1: Local preprocessing and feature extraction

Each node processes its own meter data and constructs a feature vector describing possible anomalies. The features combine:

- physical consistency residuals between measured energy E and the estimate $VI\,\mathrm{PF}\,\Delta t$;
- deviations from the meter's own historical behaviour;
- deviations from peer meters on the same feeder;
- the feeder energy-balance gap;
- indicators of stuck or repeated values;
- calendar and weather context.

Since feature scales and distributions may differ between nodes, each node standardises feature j using its own median and median absolute deviation (MAD), and clips the result to a bounded interval:

$$z_{ij} = \mathrm{clip}\left(\frac{x_{ij} - \mathrm{med}_j}{1.4826\,\mathrm{MAD}_j}, -c, c\right), \qquad c = 5. \tag{8}$$

Here, x_{ij} is feature j for record i; med_j is the local median of feature j; MAD_j is its median absolute deviation; and $\mathrm{clip}(u, -c, c)$ restricts u to the interval $[-c, c]$. The factor 1.4826 is a conventional scaling factor for the MAD. The operation limits the influence of extreme feature values on subsequent shared statistics.

Two kinds of local records are kept apart DESIGN. Records that the local trust layer has flagged but no one has confirmed are excluded from everything that is shared, so an unverified anomaly can neither distort the shared picture of normal behaviour nor be presented as an attack. Records whose cause is confirmed (by an operator, or in the prototype by the dataset labels `gt_*`) are the *only* anomalies used for the "untrusted" and family centroids of C1 and for the screening model of C3. This is a documented design rule and should not be interpreted as proof that the complete multi-node exclusion mechanism has been deployed.

8.5.3 Step 2: Prepare the information to be shared

After local preprocessing, a node prepares the relevant artifact for the selected sharing channel. The four channels serve different purposes and must not be treated as interchangeable.

C1: Anomaly signatures. C1 shares compact representations of anomaly patterns, using centroids of trusted and untrusted records in the standardised feature space. When enough examples are available, centroids may also be maintained for anomaly families such as physical bypass, data-only false data injection (FDI), and sensor drift.

Before release, centroids use feature clipping and Gaussian noise; the coordinator combines the received summaries using coordinate-wise medians. Count bucketing is also specified as a safeguard. This allows nodes to learn from patterns observed elsewhere without exchanging individual meter time series.

C2: Calibration insights. C2 shares a robust summary of the calibration behaviour of a meter model. For node k and meter model m, the summary is released only when at least three meters of that model are available locally, $n_{k,m} \geq 3$:

$$\tilde{r}_{k,m} = \mathrm{med}_{i \in m}\,\mathrm{med}_t \log\left(\frac{E_{it}}{V_{it} I_{it}\,\mathrm{PF}_{it}\,\Delta t/1000}\right). \tag{9}$$

Here, E_{it} is the measured energy for meter i at time t; V_{it} is voltage; I_{it} is current; PF_{it} is power factor; and Δt is the measurement interval (0.5 h; the factor 1000 converts W·h to kWh). Only clean readings with $V > 0$, no PV export and $E > 0$ are used: while a home exports, its imported

energy is 0 and the logarithm is undefined. The inner median summarises the time-dependent residual for each meter, while the outer median summarises the meters belonging to model m.

The coordinator then takes the median of the summaries received from eligible nodes:

$$\widetilde{r}_m = \mathrm{med}_k\, \widetilde{r}_{k,m}. \tag{10}$$

A value consistently far from zero across several nodes may indicate a systematic bias associated with that meter model. A deviation observed for only one meter is more consistent with an individual meter fault. This is a diagnostic indication, not by itself proof of the cause.

On our dataset C2 is a negative control. The three meter models differ only by communication technology (GPRS, NB-IoT, PLC), all meters have accuracy class 1.0, and no model-wide bias was injected. Computed on the clean data, $\widetilde{r}_{k,m}$ lies within ± 0.0002 (i.e. $\pm 0.02\%$) for every node and model, so any larger value would be a false alarm. With the $n_{k,m} \geq 3$ rule, the PLC model (5 meters in N1, none in N2, 2 in N3) is released by N1 only, so its "median across nodes" rests on a single node. A positive test is planned (Section 8.8): a +3% gain is injected on all NB-IoT meters *inside the experiment code*, leaving the dataset unchanged; C2 must report it, while a single biased meter (`bias_step`) must not move the model-level value.

C3: Federated model updates. C3 allows nodes to contribute to a shared screening model without transferring their local training records. At round t, node k trains the current model w_t on its local dataset $\mathcal{D}_k$. The resulting update is

$$\Delta_k = \mathrm{LocalTrain}(w_t; \mathcal{D}_k) - w_t. \tag{11}$$

The update Δ_k represents the change produced by local training. It is not the raw dataset and does not need to contain individual meter readings.

C4: Urgent alerts. C4 is a separate proposed path for rare, urgent information. An alert may contain a signed attack fingerprint describing the subtype family, feature-residual profile, affected meter model, and severity. The design requires confirmation by a human operator at the source, rate limiting, and expiration. Receiving nodes should treat the alert as a prior indication to investigate, not as a definitive verdict. The full distributed alert path remains a design requirement.

The dataset contains a ready-made case. A `coordinated_fdi` attack (three meters of one feeder falsified together) hits N3 from 1 to 28 July 2025 and N1 from 14 October 2025. Once N3's operator confirms the July attack, its fingerprint reaches N1 three months before N1's own attack. Because Goal 1's physics rule already flags data-only injection in N1 from the first day, the expected benefit of the alert is not earlier detection but a faster and more confident diagnosis: the receiving node can name a confirmed attack family instead of an unexplained anomaly.

8.5.4 Step 3: Protect model updates before sharing

Before a C3 update is shared, the node clips its Euclidean norm and adds Gaussian noise:

$$\widetilde{\Delta}_k = \Delta_k \min\left(1, \frac{C}{\|\Delta_k\|_2}\right) + \mathcal{N}\left(0, \frac{\sigma^2 C^2}{K} I\right). \tag{12}$$

The symbols have the following meanings:

- C is the maximum allowed update norm;
- $\|\Delta_k\|_2$ is the Euclidean norm of the local update;
- σ is the noise multiplier;
- K is the number of participating nodes;
- I is the identity matrix;

- $\mathcal{N}(0, \frac{\sigma^2 C^2}{K} I)$ is the zero-mean Gaussian noise distribution used by the design.

If the update norm is no greater than C, the clipping factor is one. If it exceeds C, the factor reduces the update until its norm is C. Gaussian noise is then added as a privacy safeguard.

Clipping limits the influence of a single large update, while noise is intended to reduce the information revealed by the shared update. However, clipping and adding noise alone do not establish a formal differential-privacy guarantee. Such a guarantee would require a specified privacy analysis and accounting procedure.

8.5.5 Step 4: Validate and aggregate contributions

The coordinator first checks received contributions against the announced protocol version and feature schema. It also checks freshness, update norms, and replay nonces, discarding invalid contributions. The protocol requires authenticated and encrypted communication; secure aggregation remains a deployment design requirement.

For valid C3 updates, the coordinator applies the selected aggregation rule and updates the shared model:

$$w_{t+1} = w_t + \mathrm{Agg}\Big(\widetilde{\Delta}_1, \ldots, \widetilde{\Delta}_K\Big). \tag{13}$$

Here, Agg is the aggregation function, and w_{t+1} is the candidate shared model for the next round. Three aggregation rules are considered.

FedAvg. FedAvg takes the arithmetic mean of the protected updates:

$$\mathrm{Agg}_{\mathrm{FedAvg}} = \frac{1}{K} \sum_{k=1}^{K} \widetilde{\Delta}_k. \tag{14}$$

This provides a simple way to combine the contributions of all participating nodes. However, an unusually large or malicious update may still affect the average, which motivates robust alternatives.

Coordinate-wise median. The coordinator calculates the median separately for each model coordinate j:

$$(\mathrm{Agg}_{\mathrm{med}})_j = \mathrm{med}_k \left(\widetilde{\Delta}_{k,j}\right). \tag{15}$$

Compared with the mean, the coordinate-wise median is less sensitive to extreme values in individual coordinates. Its robustness has limits, especially when few nodes participate or several nodes collude.

Reputation-weighted aggregation. The coordinator can also weight updates according to node reputation and how far each update deviates from the median update:

$$\mathrm{Agg}_{\mathrm{rep}} = \sum_{k=1}^{K} \rho_k \widetilde{\Delta}_k, \qquad \rho_k \propto r_k \exp\Big(-\frac{d_k}{\tau}\Big), \tag{16}$$

where

$$d_k = 1 - \cos\Big(\widetilde{\Delta}_k, \, \mathrm{med}_{k'} \, \widetilde{\Delta}_{k'}\Big). \tag{17}$$

In this expression, $r_k \in [0, 1]$ is the reputation of node k, d_k measures the cosine distance between its update and the coordinate-wise median update, and τ controls the effect of that distance on the weight. The proportional weights are normalised so that they sum to one. Reputation is designed to decrease after rejected rounds and recover after accepted rounds.

The three aggregation rules are alternatives; they are not three successive operations that must all be applied to the same update. The choice of aggregation rule should be stated for each experiment or deployment configuration.

For C1 anomaly centroids, the coordinator instead applies coordinate-wise medians to the protected summaries. C2 calibration summaries are combined using the median across eligible nodes, as defined in Equation 10.

8.5.6 Step 5: Publish the shared artifact

After validation and aggregation, the coordinator publishes a signed, versioned artifact containing the relevant shared result. The coordinator announces the round identifier, protocol version, feature-schema identifier, clipping and noise parameters (c, C, σ), and deadline. Nodes sign and submit their contributions through the specified authenticated, encrypted channel.

Versioning allows nodes to identify which shared artifact they have received. Signatures support integrity and origin verification, while the validation checks help reject stale or malformed contributions. These protections describe the intended protocol; they do not imply that a complete secure multi-node transport and aggregation system has been implemented.

8.5.7 Step 6: Local canary evaluation and adoption

A node does not automatically adopt a newly published model. It first evaluates the candidate on held-out local canary data and checks the following criteria:

- ROC-AUC must not decrease beyond an allowed margin Δ_{AUC};
- the false-positive rate must remain within the defined alarm budget;
- genuine `grid_event` and `equipment_failure` records must not be misclassified as data-quality problems;
- recall per attack *family* (physical bypass, data-only FDI, sensor drift, corruption) must not decrease. Per-type recall cannot serve as a gate: each falsification type occurs at most once per node, so its recall can only be 0 or 1. A family gates adoption only where the node holds at least three events of it (Table 25); otherwise its result is reported to the operator without blocking adoption.

Table 25. Events per attack family and node, and whether the family can gate adoption (≥ 3 events).

Family	N1	N2	N3
Physical bypass	3 (gate)	2	2
Data-only FDI	5 (gate)	2	3 (gate)
Sensor drift	8 (gate)	6 (gate)	4 (gate)
Corruption	25 (gate)	14 (gate)	14 (gate)

The numerical thresholds for these criteria must be specified before deployment. If the candidate passes the checks and receives the required operator approval, the node adopts it. Otherwise, the node retains its previous model and logs the rejection.

This local evaluation step limits the risk of adopting a shared model that is less suitable for a particular neighborhood. It does not, by itself, prove that the model is safe against every attack or that the protocol improves detection performance.

8.5.8 Implementation status and limitations

The protocol combines local processing, protected information sharing, robust aggregation, and local model evaluation. Its claims must nevertheless be interpreted in light of the current implementation scope:

- The experimental data are synthetic; they are not real STEG meter readings.

- The complete multi-node sharing experiment has not been run. The current work does not establish a measured performance improvement from cross-node collaboration.
- Network transport, node authentication, and secure aggregation remain documented design requirements.
- The noise mechanisms described here should not be presented as a formal differential-privacy guarantee without a complete privacy analysis.
- Robust aggregation has limitations with a small number of participants. With only three nodes, for example, a coordinate-wise median cannot guarantee protection against coordinated malicious contributions.

Accordingly, this section specifies the proposed shared-learning protocol and its safeguards. What exists as code today are the protocol primitives in `src/energymind/shared/`: update clipping and Gaussian noise (Eq. 12), FedAvg and coordinate-wise median aggregation, and the C2 calibration summary. The multi-node rounds, the coordinator, transport security and secure aggregation are design.

8.6 Privacy: what is protected and what is not

Layered protections, each with a distinct role: locality (no raw data), clipping (bounded influence), Gaussian noise (plausible deniability for individual contributions [34, 35]), group suppression ($k \geq 3$, in the spirit of k-anonymity [28], *not* a DP guarantee) and secure aggregation so the coordinator sees only the sum [33]. Gradient-style updates can leak training data if sent in the clear [37], which is why secure aggregation is part of the deployment design.

For FedAvg with all K nodes participating in each of R rounds, Gaussian noise multiplier σ and clipping bound C, the Rényi-DP accounting of [36] gives

$$\varepsilon(\delta) = \min_{\alpha>1} \left[\frac{R\alpha}{2\sigma^2} + \frac{\ln(1/\delta)}{\alpha-1} \right] = \frac{R}{2\sigma^2} + \sqrt{\frac{2R\ln(1/\delta)}{\sigma^2}}, \qquad \delta = 10^{-5}. \tag{18}$$

This assumes the summed noise (each node adds $\sigma^2 C^2/K$) is observed only after secure aggregation. Table 26 evaluates Eq. (18); the values are computed from the formula, not measured on our data.

Table 26. ε at $\delta = 10^{-5}$ from Eq. (18) (node-level, FedAvg only).

σ	$R = 10$	$R = 26$	$R = 52$
1	20.2	37.5	60.6
2	8.8	15.5	23.8
4	4.1	6.9	10.3
8	2.0	3.3	4.7

Honest reading. Even a weekly schedule ($R = 52$) needs $\sigma \approx 8$ for $\varepsilon < 5$, and with $K = 3$ that much noise will likely cost accuracy, which is exactly the privacy–utility trade-off the planned experiment will measure (Section 8.8); no utility number is claimed in Phase 1. Eq. (18) covers FedAvg only; it does not cover signatures, calibration summaries, class counts, participation metadata, or the median and reputation rules, which need their own analysis. Until secure aggregation and transport security are implemented and the accounting is reviewed, the system is *privacy-aware*, not *provably private*.

8.7 Threat model and collective defence

Table 27. Threat–defence matrix and what remains uncovered.

Threat	Example	Defence	Residual risk
Honest-but-curious coordinator	Inspects each node's update/summary	Secure aggregation, noise, bucketed counts	Not implemented (design)
Network eavesdropper / tamperer	Reads or alters artifacts in transit	TLS, signed and versioned artifacts	Key management out of scope
Curious peer node	Infers another node's behaviour from the shared model	Clipping, noise, $k \geq 3$ suppression	No formal bound beyond Eq. (18)
Poisoned update	Sign-flipped or oversized Δ	Norm clip, median / reputation, canary test	Tolerates at most 1 of 3 arbitrary outliers
Replay / stale update	Resends an old Δ	Round nonce, freshness check	—
Backdoor / adaptive attack	Update crafted to blind the model to one subtype [38]	Canary per-family recall, human approval, rollback	Not proven robust; basic defences are known to be weak
False alert injection (C4)	Fake "attack fingerprint" to cause alarms or desensitise	Source-node human confirmation, signature, rate limit, expiry, prior-only use	Sybil nodes need controlled registration
Collusion / Sybil	Majority of nodes (or fake identities) collude	Controlled node registration, governance rules	**Not defended**: median fails if ≥ 2 of 3 collude

Over-trust and under-trust of shared knowledge. The Goal 3 concern about operators applies to the network as well. *Over-trust* (blindly adopting a shared model) is prevented by the canary test, per-family recall checks, human approval and rollback. *Under-trust* (rejecting useful updates forever) is monitored: every rejection is logged with its reason and the shadow-model comparison, so a persistent pattern of rejected-but-beneficial updates is visible to reviewers. Shared outputs only adjust anomaly scores; they do not change the autonomy thresholds or trigger any action on the grid, and Goal 3's non-tunable routing rules (R0–R3) apply whatever is shared.

8.8 Evaluation plan

No experiment across nodes has been run yet, and no performance number for sharing is claimed. The planned evaluation, all on the existing dataset (no change to the data), is:

1. **Splits.** A calendar split does not work: in N1, November–December contain no falsification and no sensor-bias event. Models are therefore evaluated by cross-validation with folds grouped by event (`gt_event_id`), as in Goals 1–2; results are reported per event, with intervals. The `gt_*` columns are never model inputs.
2. **Transfer to unseen attacks.** For every attack type that is missing in some node (Table 22), train on the nodes that have it and score its events while the receiving node's own data contain no example of it; compare local only, local + C1 centroids, and local + C3 (FedAvg vs. coordinate-wise median). Measure per-family recall at a fixed false-positive rate.
3. **Privacy–utility.** Repeat the C3 run for $\sigma \in \{1, 2, 4, 8\}$ and plot detection against the ε of Table 26.
4. **Corrupted node.** One node sends sign-flipped or scaled updates; compare FedAvg, median and reputation-weighted aggregation, and check that the canary rejects the poisoned artifact.

5. **C2.** Negative control on the clean data (all models ≈ 0), then the +3% NB-IoT injection done in code.
6. **C4.** N3's July coordinated attack as a prior for N1 before 14 October: diagnosis time and confidence with and without the alert.

8.9 Limitations

1. All data and anomalies are synthetic; results may reflect the simulator and cannot establish real-grid performance. External validation (e.g. Low Carbon London, UCI) is needed.
2. The three nodes come from one generator, so their heterogeneity understates that of independent real operators.
3. With $K = 3$, robust aggregation has almost no redundancy and strong DP is costly (Table 26).
4. The planned corrupted-node test (Section 8.8) covers sign-flipped and scaled updates only, not backdoors, collusion, adaptive attacks, Sybil identities or a compromised coordinator.
5. The simulated evasion attack is a random-walk scaling, not the gradient-based attacks of the literature.
6. Each falsification type occurs 2–3 times in the dataset and at most once per node, so per-type results are case studies; families are used for decisions, with wide uncertainty.

9 Business Scope: Deployment, Integration, Continuous Training and Budget

What is sold and to whom. A software analytics layer for a distribution utility (STEG) that sits on top of the smart-meter infrastructure it already operates: no new device at the customer. It turns the half-hourly meter data into verdicts that an operator can trust and act on. Its value drivers are fewer wasted field visits (a genuine grid event is not sent out as a "faulty meter"), earlier detection of meter manipulation, correct billing when a meter drifts, and operator time spent only where it matters. These will be quantified in Phase 2 on the simulated nodes; no saving is claimed here.

9.1 Deployment model

The system runs centrally, next to the utility's meter data management system (MDMS): Goals 1–3 run per node every day (every 30 minutes for urgent rules such as supply loss), Goal 4's coordinator runs once a week. It needs, per neighbourhood: the existing smart meters, one accurate meter at the head of each low-voltage feeder, and the operator console (a web application). Roll-out is staged, and each stage must pass its gate before the next starts:

Table 28. Staged deployment.

Stage	Duration	What runs	Gate to the next stage
A. Shadow pilot, 1 neighbourhood	3 months	Goals 1–3 produce verdicts and routes, but nothing is acted on; verdicts are compared with field inspections	False positives within the alarm budget; zero unsafe automatic actions; operators judge explanations useful
B. Live pilot, 3 neighbourhoods	6 months	Operator workflow live (G0–G3); Goal 4 sharing between the three nodes	Missed-escalation rate below target; canary adoption working; privacy review passed
C. Scale-up, district / region	after Stage B, no fixed end	More nodes join; Goal 4 gains participants, which also strengthens robust aggregation	Review load per operator stays within capacity

9.2 Integration

Table 29. Interfaces with the utility's systems. Integration is the main one-off cost.

Interface	Content	Mechanism
Meter data in	30-minute interval readings and meter event logs	MDMS export (IEC 61968-9 / CIM messages, or scheduled batch files)
Feeder-head data in	Feeder energy and voltage	MDMS or SCADA; a feeder-head meter is installed where none exists
Context in	Temperature, irradiance, calendar	National weather service feed; holiday and Ramadan calendar
Outage management (out)	Genuine supply-loss events	Event message to the OMS
Field service (out)	Inspection requests for suspected meters	Work order in the field-service system
Billing (out)	Hold or estimate for readings judged untrustworthy	Flag in the billing workflow (only after G2b/G3 decisions)
Identity and audit	Operator roles, two-person approval, hash-chained log	Utility single sign-on; log kept in the utility's archive

9.3 Continuous training

- **Daily, automatic:** each meter's 28-day profile and each feeder's balance reference move forward.
- **Monthly:** thresholds and the confidence calibration are refitted on newly confirmed cases (field inspection results, operator closures, blind operator judgements), with the event-grouped validation of Section 3. A new version is promoted only after a shadow comparison with the current one, and every version (`configs/thresholds.yaml`) can be rolled back.
- **Quarterly:** the learned detectors (Isolation Forest, One-Class SVM, autoencoder) are retrained on the latest clean data.
- **Weekly:** Goal 4 sharing rounds; each shared artifact passes the local canary test and operator approval before adoption.
- **Monitoring, continuously:** false positives against the alarm budget, calibration (Brier score per class), share of reviewed cases per operator-hour, and alignment between AI and operator confidence; a breach triggers an early refit or a rollback.

9.4 Budget

What we measured. Compute is not a cost driver. The complete Goal 1 experiment on one neighbourhood for a whole year (30 meters, 525 600 readings, five-fold training of six detectors) runs in about 30 seconds on a laptop. Daily operation scores only one day per node, so a single ordinary server can serve many neighbourhoods; explanations are generated only for flagged meter-days. The real operating cost is **operator review time**: at the current false-positive rate (2.3% of clean meter-days, Section 5), a 30-meter neighbourhood produces about 250 false-alarm meter-days a year, most of which the routing keeps at G0–G1; reducing this is a Phase 2 objective.

Pilot budget (stages A and B, 9 months). Effort is given in person-months; the unit costs are *planning assumptions* to be validated with STEG, not quotes, and the total scales linearly with them.

Table 30. Indicative pilot budget. Unit costs are planning assumptions (TND = Tunisian dinar).

Item	Quantity	Assumed unit cost	Total (TND)
Data / ML engineers (Goals 1–4, retraining)	2 × 9 person-months	5 000 / person-month	90 000
Integration engineer (MDMS, OMS, field service, billing)	6 person-months	5 000 / person-month	30 000
Distribution-grid expert (rules, scenarios, operator training)	3 person-months	5 000 / person-month	15 000
Feeder-head meters where missing (3 nodes, 7 feeders)	up to 7	2 500 installed	up to 17 500
Server (central, one virtual machine)	9 months	500 / month	4 500
Operator time for pilot reviews and training	2 operators, part time	internal	internal
Total (excluding internal operator time)			**about 157 000**

Scale-up costs. Integration is paid once; each new neighbourhood then adds mainly feeder-head meters where missing, a short shadow period, and operator review time. The language model for explanations can run locally (no per-call fee) or as a service; in both cases only flagged meter-days are explained, and the plain evidence summary is always available as a free fallback.

10 Innovation

- **Physics as the explanation.** The strongest detectors are physical laws a technician can check on site: energy must match $V\,I\,\mathrm{PF}\,\Delta t$, a meter cannot read more voltage than its feeder head, the feeder must balance. Machine learning adds sensitivity, not authority.
- **Grid event versus bad data as the central question**, answered with cross-sensor evidence rather than a single anomaly score.
- **Confidence that knows its limits**: a class whose calibration has not been validated cannot be shown as highly confident, and confidence can never unlock a consequential action.
- **Blind-first human review**, so the operator's judgement is independent evidence rather than an echo of the AI.
- **Honest evaluation**: event-grouped validation, per-event reporting, and explicit listing of what the system still misses.

11 Phase 2 Work Plan (to 12 December 2026)

Period	Work
16 Oct – 8 Nov	Goal 1: feeder-balance residual model and meter attribution; Goal 2: fit the cascade thresholds and calibration out-of-fold, SHAP, verdict cards
9 Nov – 22 Nov	Goal 3: governance engine integrated with Goals 1–2, reference scenarios, synthetic operators, coverage sweep; operator console
23 Nov – 2 Dec	Goal 4: three-node simulation (transfer, privacy–utility, corrupted node, C2, C4)
3 Dec – 12 Dec	End-to-end demo on N1, full report, video and presentation

Repository

The repository contains the dataset (`data/`), the data card, the four goal design reports (`docs/goal1_detection` ... `docs/goal4_shared_learning`), the decision-log schema, the configuration files (thresholds, asset overlay), the Python package (`src/energymind`: data loading and label hygiene, rules, verdict cascade, hash-chained decision log, the Version 1 escalation engine, sharing primitives), the Goal 1 experiment that produces every detection number quoted here (`docs/goal1_detection/n1_experiment.py`), the script that prints every dataset fact quoted here (`scripts/check_dataset.py`), a notebook and the tests (verdict rules, the six governance scenarios and five safety invariants on 400 random cases each). All data are simulated; no real STEG data is included.

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